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(54) **VIBRATION APPARATUS AND VEHICULAR APPARATUS COMPRISING THE SAME**

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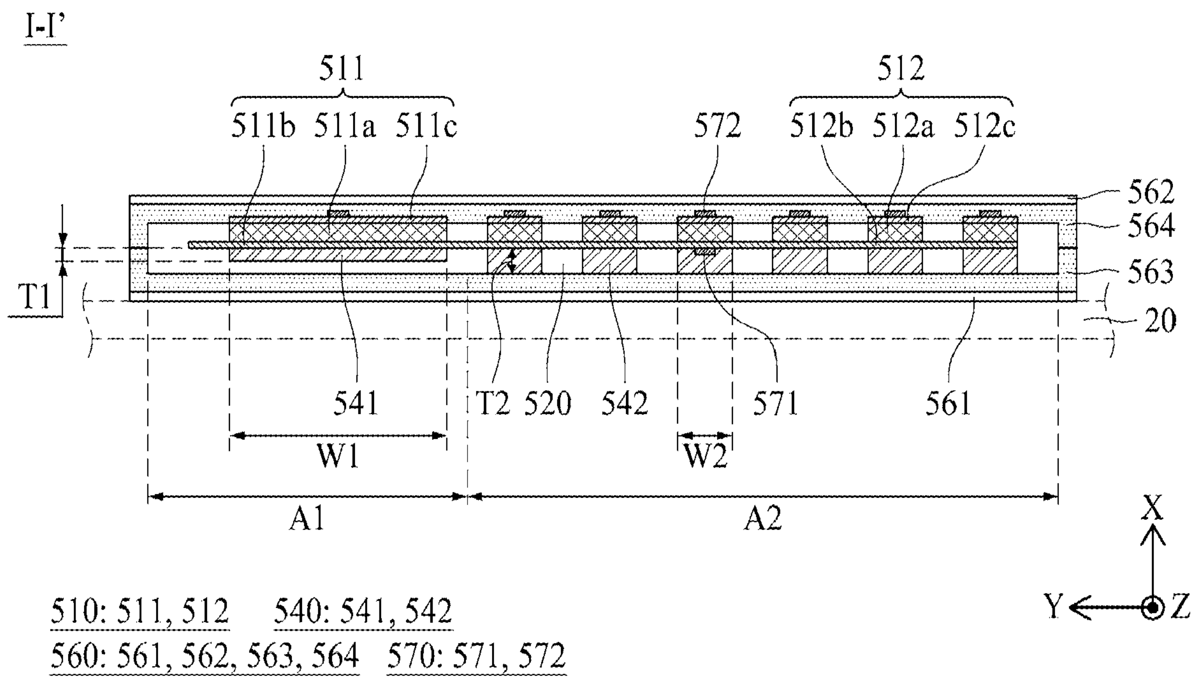
CPC ***H04R 3/12*** (2013.01); *H04R 2499/13* (2013.01)

(57) **ABSTRACT**

One or more embodiments of the present disclosure provides a vibration apparatus and a vehicular apparatus including the same. The vibration apparatus comprises a cover member having first and second areas, a vibration part in each of the first and second areas, and a support member between the cover member and the vibration part. The vibration apparatus and the vehicular apparatus including the same according to one or more embodiments of the present disclosure can prevent or minimize acoustic interference between multiple speakers, and can improve the directionality, the sound characteristics, and/or the sound pressure characteristics.



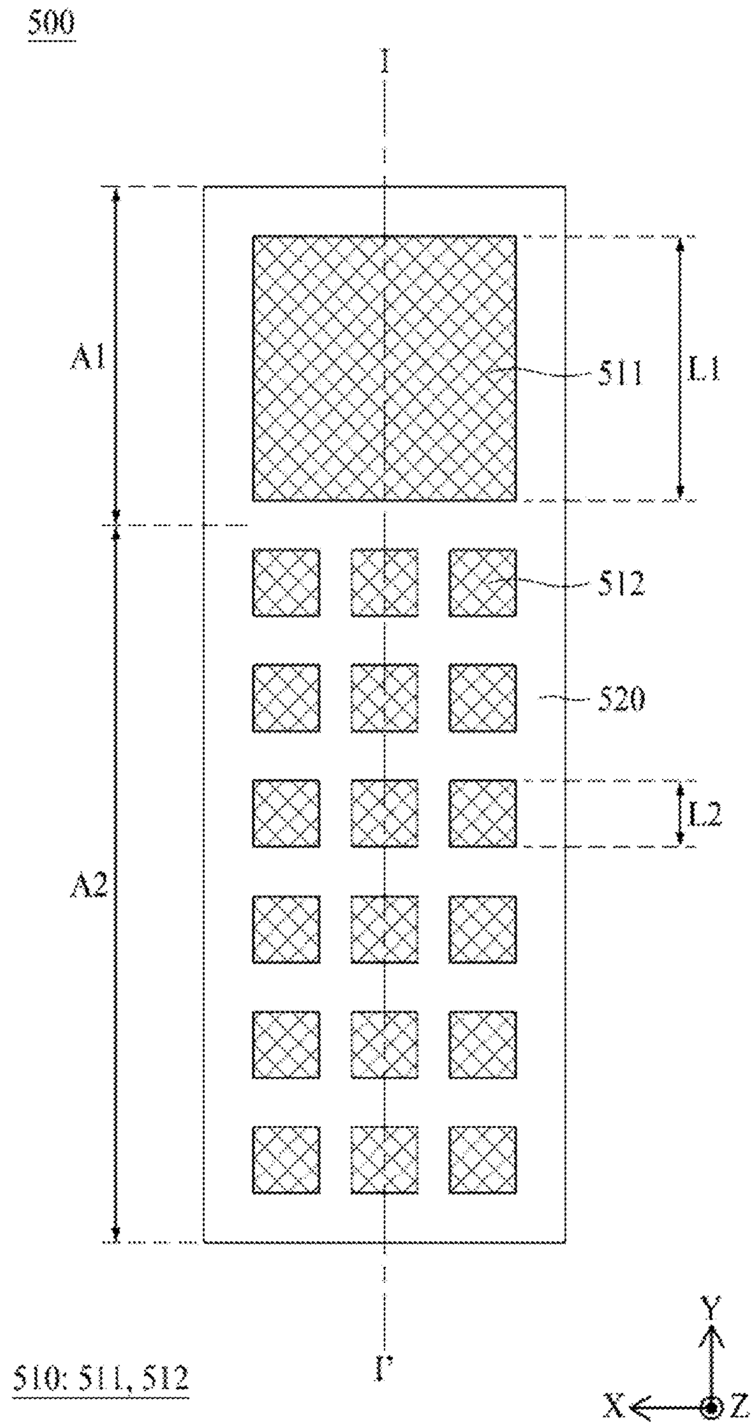
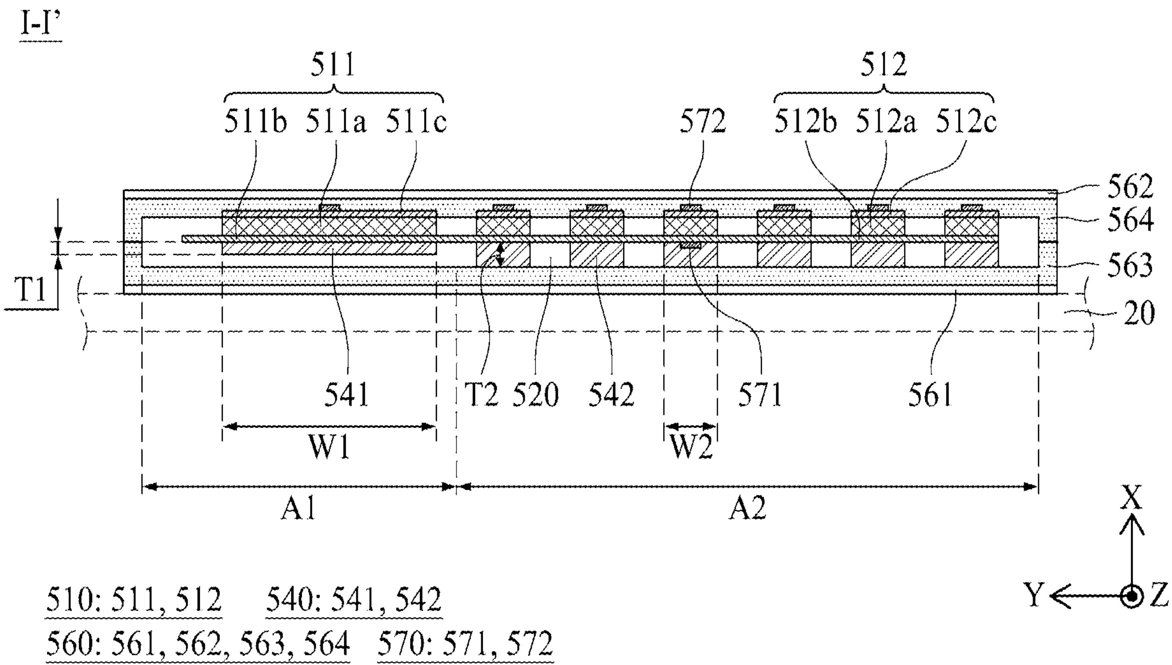


FIG. 1


I-I'
511
511b 511a 511c
572
512
512b 512a 512c
562
564
563
T1
20
541
T2 520
542
571
561
W1
W2
A1
A2
X
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Z
510: 511, 512
540: 541, 542
560: 561, 562, 563, 564
570: 571, 572

FIG. 2

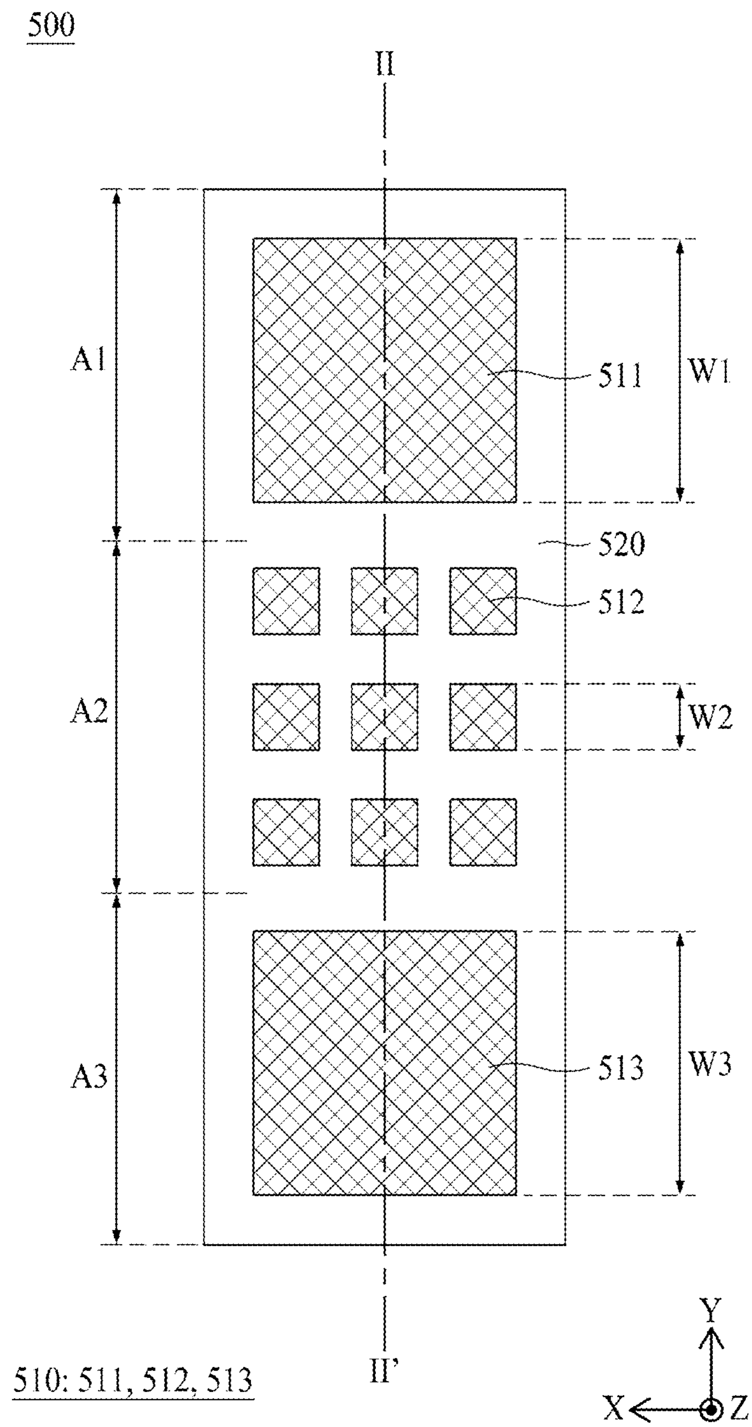
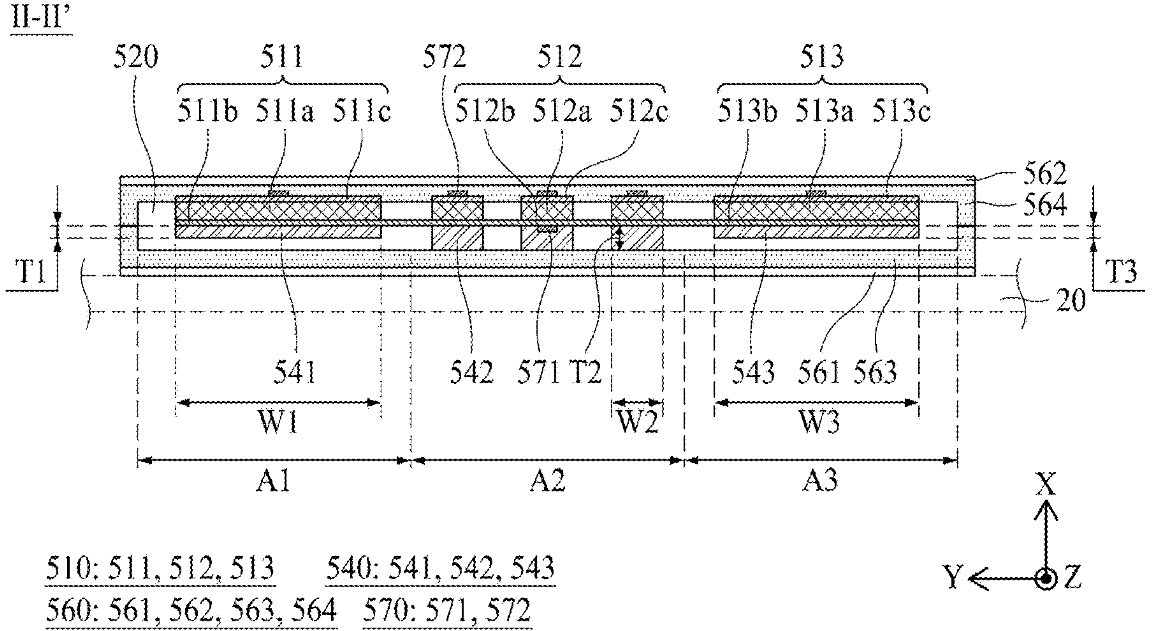


FIG. 3


II-II'
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511b 511a 511c
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513
513b 513a 513c
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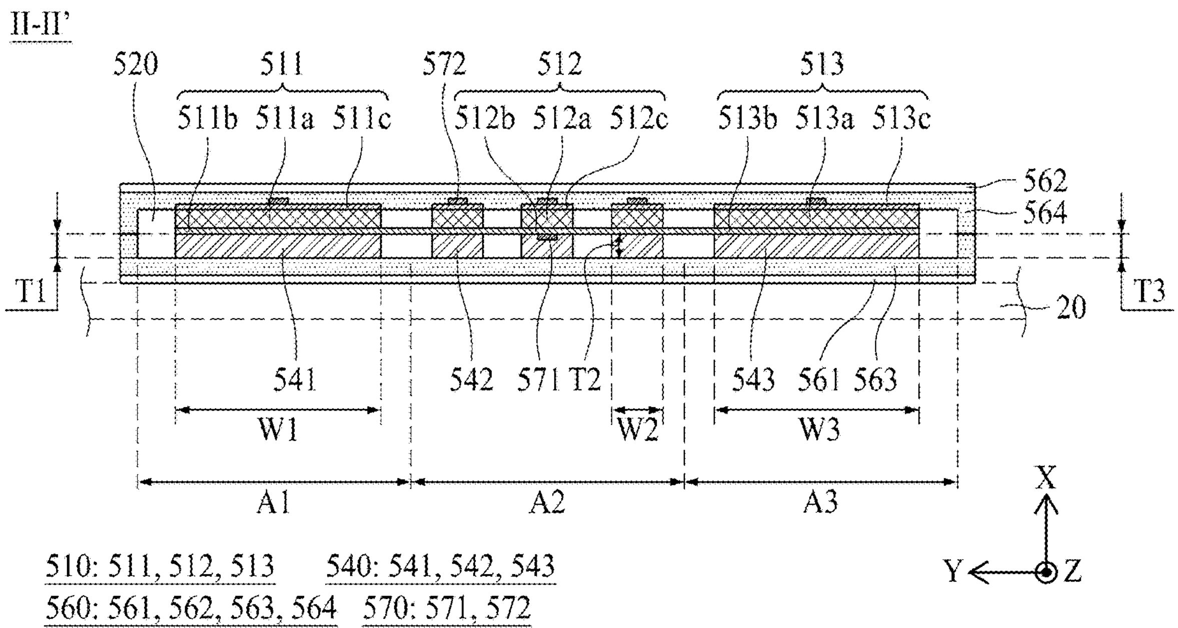
T3
20
541
542
571 T2
543 561 563
W1
W2
W3
A1
A2
A3
X
Y
Z
510: 511, 512, 513
540: 541, 542, 543
560: 561, 562, 563, 564
570: 571, 572

FIG. 4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513b 513a 513c
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541
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571 T2
543 561 563
W1
W2
W3
A1
A2
A3
X
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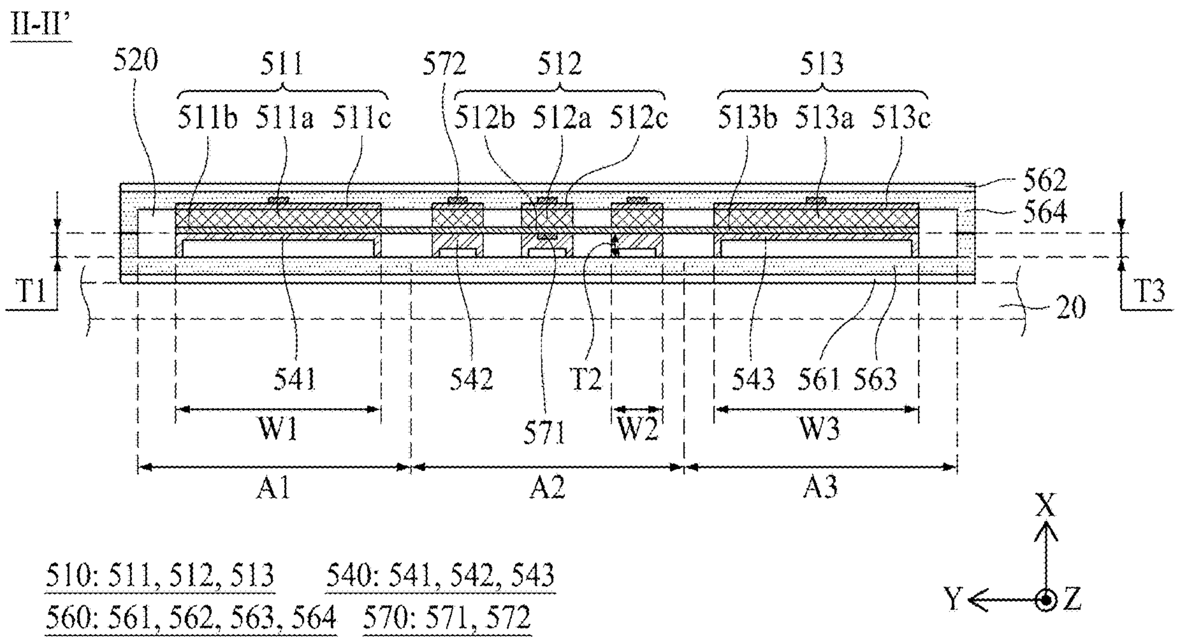
510: 511, 512, 513
540: 541, 542, 543
560: 561, 562, 563, 564
570: 571, 572

FIG. 5


II-II'
520
511
511b 511a 511c
572
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512b 512a 512c
513
513b 513a 513c
562
564
T1
T3
20
541
542
T2
543 561 563
W1
571
W2
W3
A1
A2
A3
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560: 561, 562, 563, 564
570: 571, 572
X
Y
Z

FIG. 6

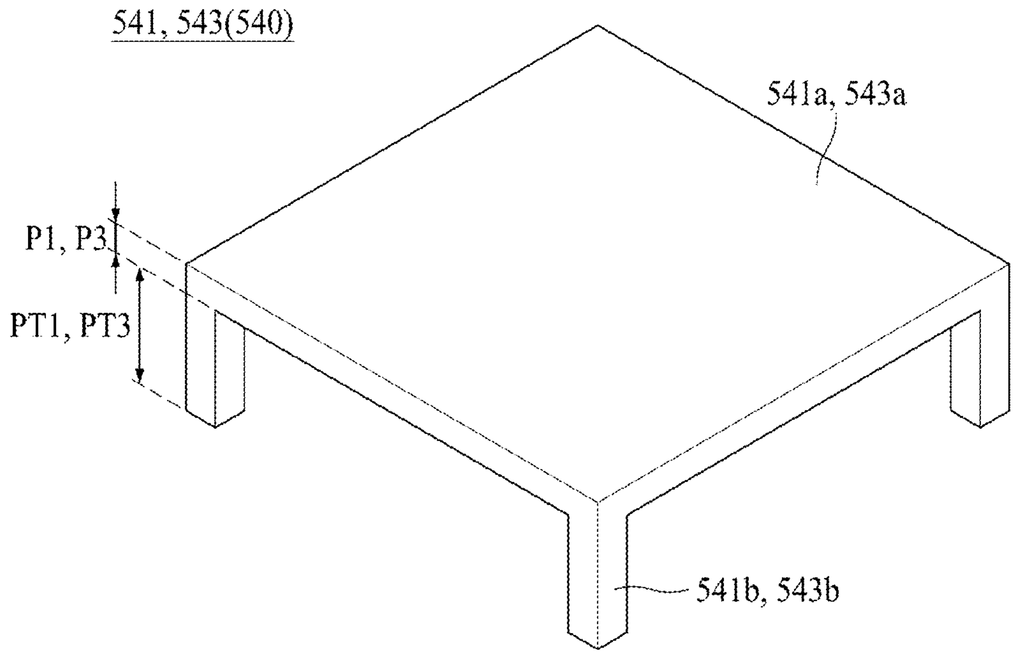

541, 543(540)
541a, 543a
P1, P3
PT1, PT3
541b, 543b

FIG. 7A

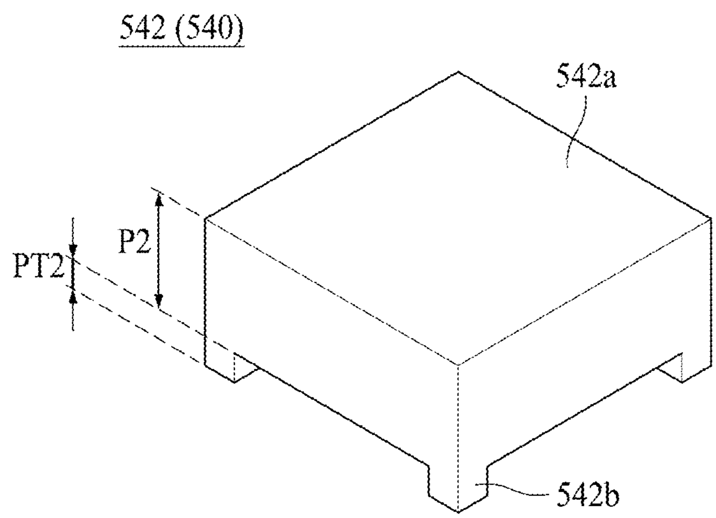

542 (540)
542a
P2
PT2
542b

FIG. 7B

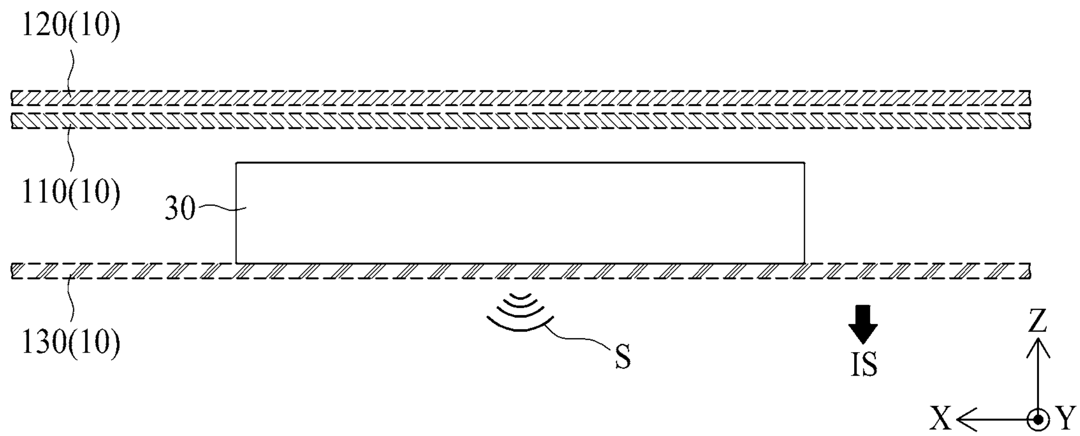

120(10)
110(10)
30
130(10)
S
IS
Z
X
Y

FIG. 8

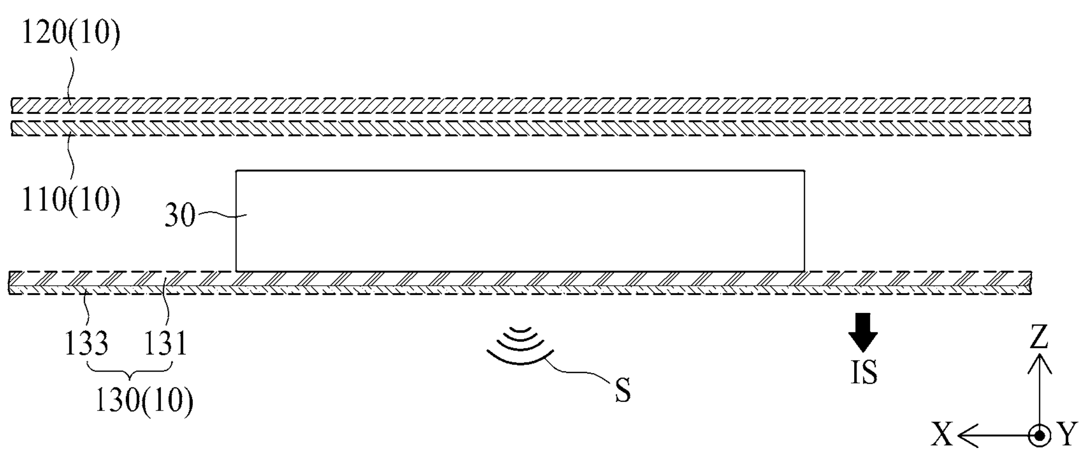

120(10)
110(10)
30
133
131
130(10)
S
IS
Z
X
Y

FIG. 9

FIG. 10

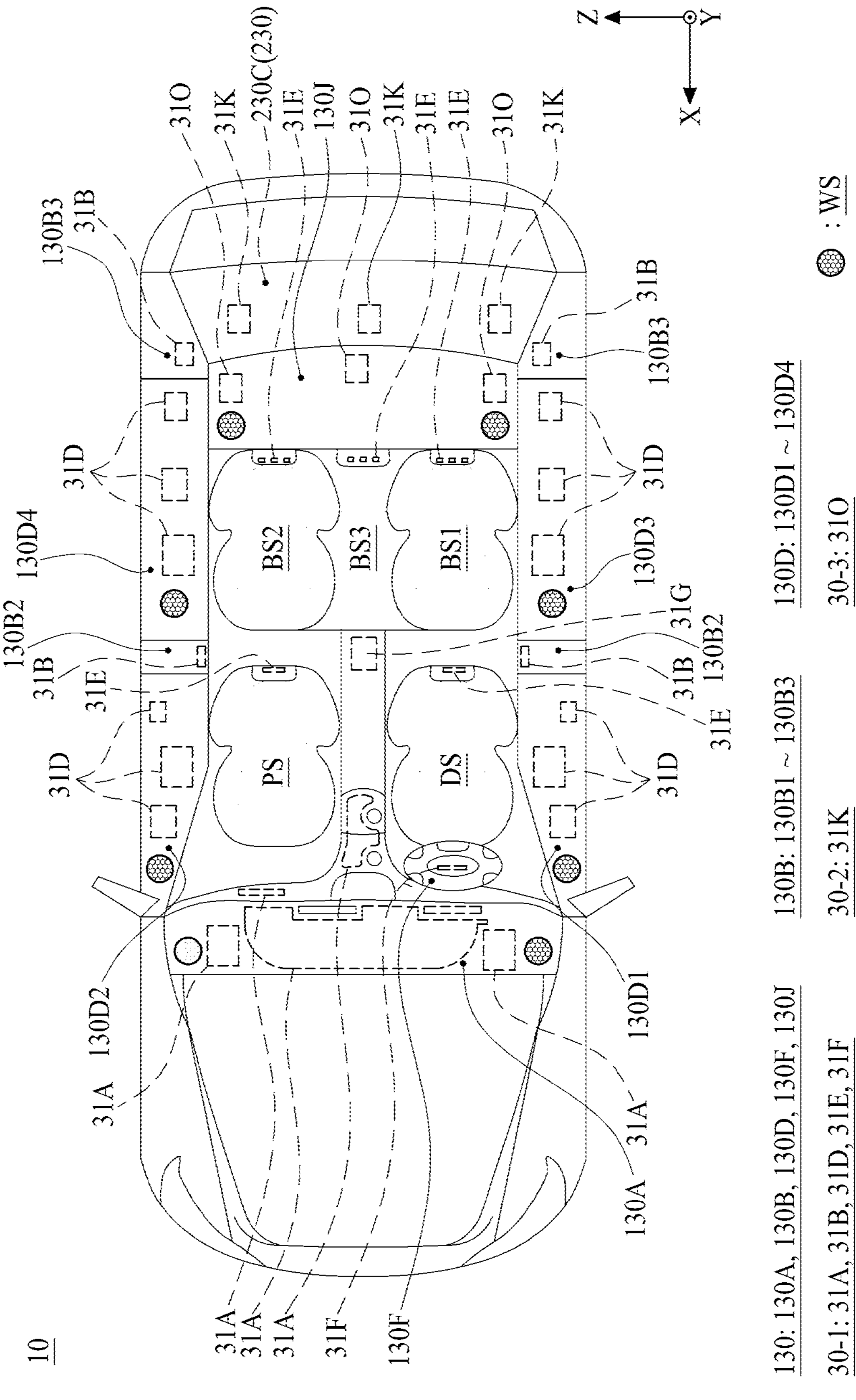
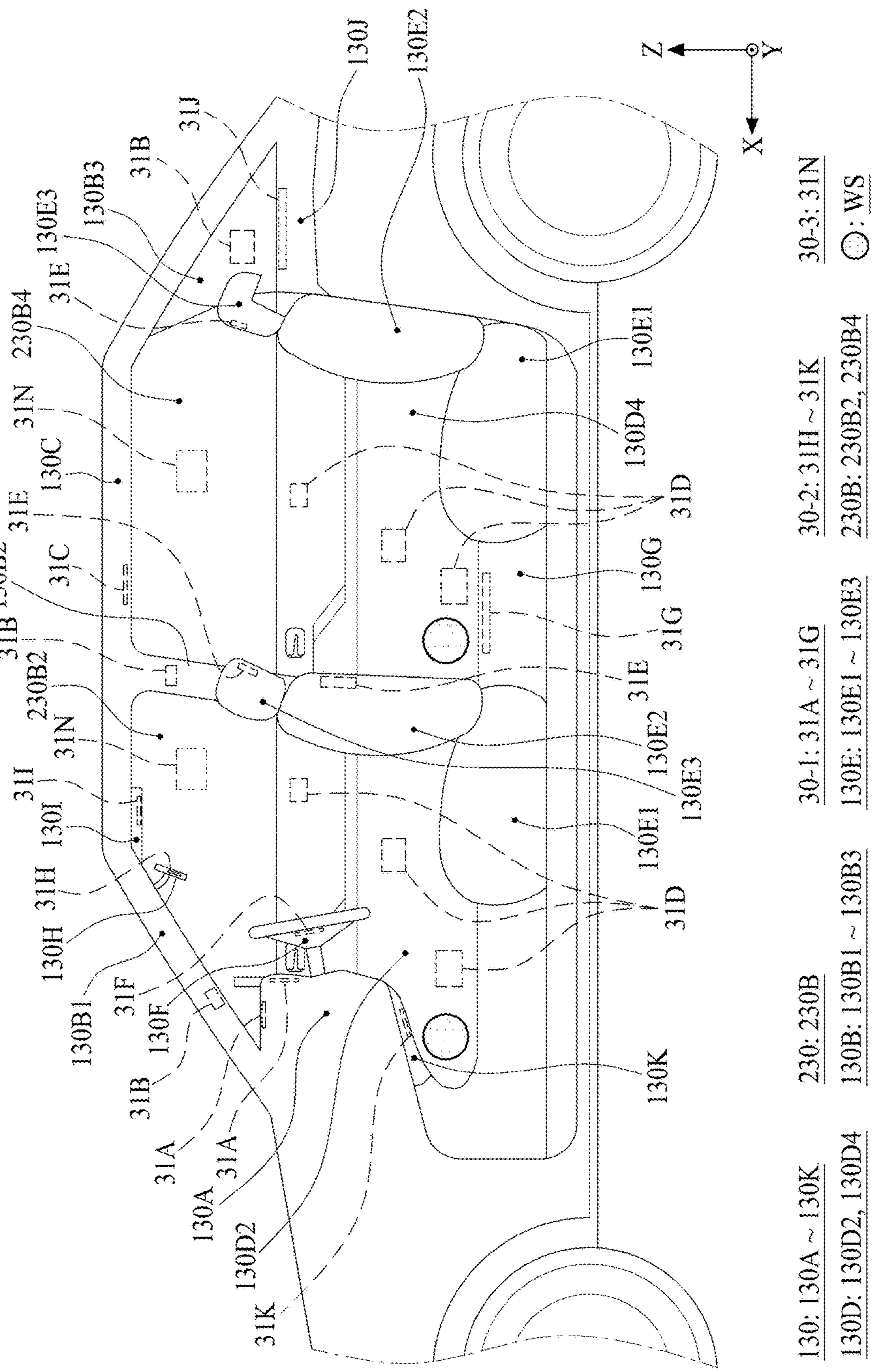

10
130B3
31B
31O
31K
230C(230)
31E
130J
31O
31K
31E
31E
31O
31K
Z
Y
X
31D
130D4
130B2
31B
31E
BS2
BS3
BS1
PS
DS
31G
130D3
130B2
31B
31E
31D
130D2
31A
130D1
31A
130A
31A
31A
31A
31F
130F
130: 130A, 130B, 130D, 130F, 130J
30-1: 31A, 31B, 31D, 31E, 31F
130B: 130B1 ~ 130B3
30-2: 31K
130D: 130D1 ~ 130D4
30-3: 31O
: WS

FIG. 11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130E2
31J
31B
130B3
130E3
31E
230B4
31N
130C
31E
31C
130B2
31B
230B2
31N
31I
130I
31H
130H
130B1
31F
130F
31B
31A
31A
130A
130D2
31K
130E1
130D4
31D
130G
31G
31E
130E2
130E3
130E1
31D
130K
Z
Y
X
130: 130A ~ 130K
130D: 130D2, 130D4
230: 230B
130B: 130B1 ~ 130B3
30-1: 31A ~ 31G
130E: 130E1 ~ 130E3
30-2: 31H ~ 31K
230B: 230B2, 230B4
30-3: 31N
: WS

FIG. 12

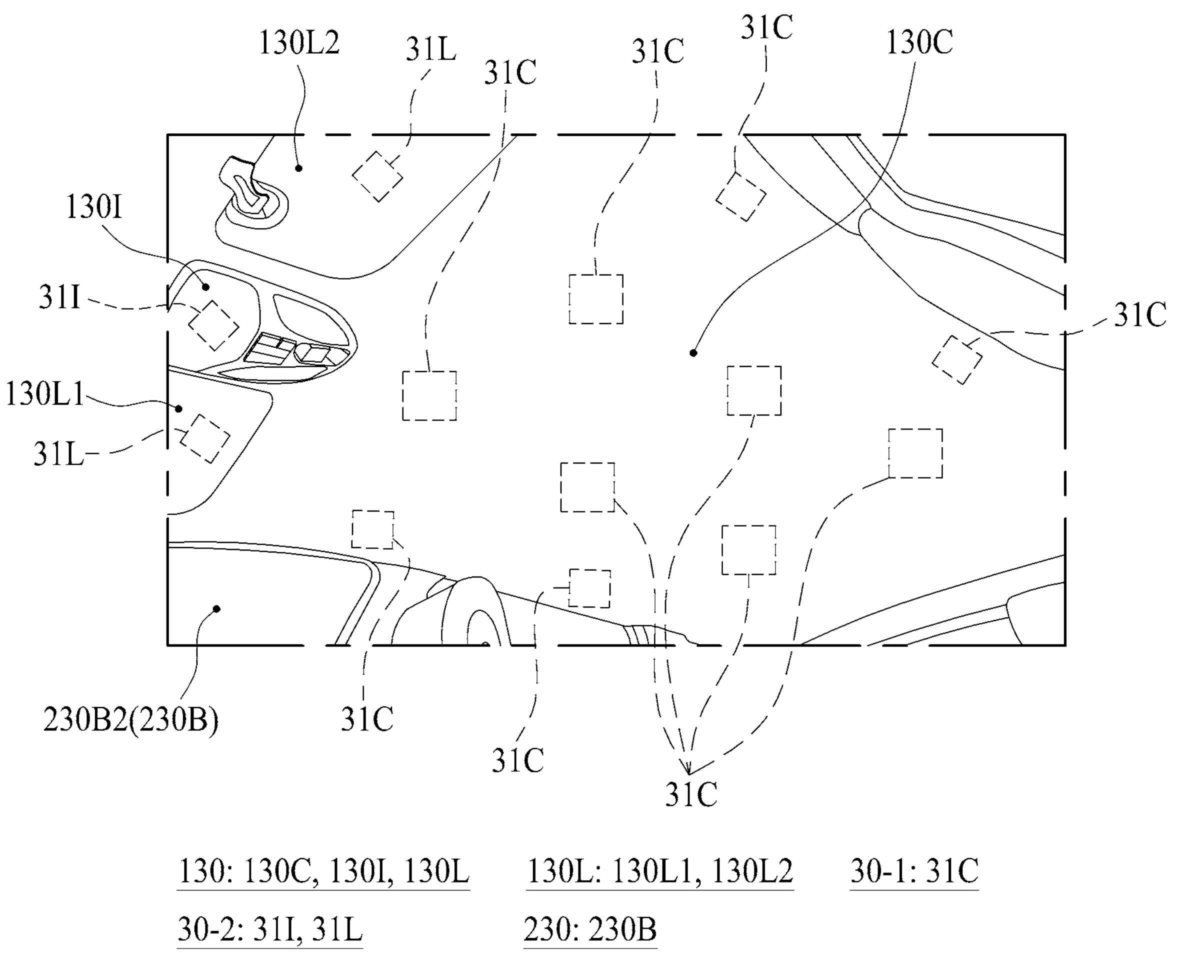
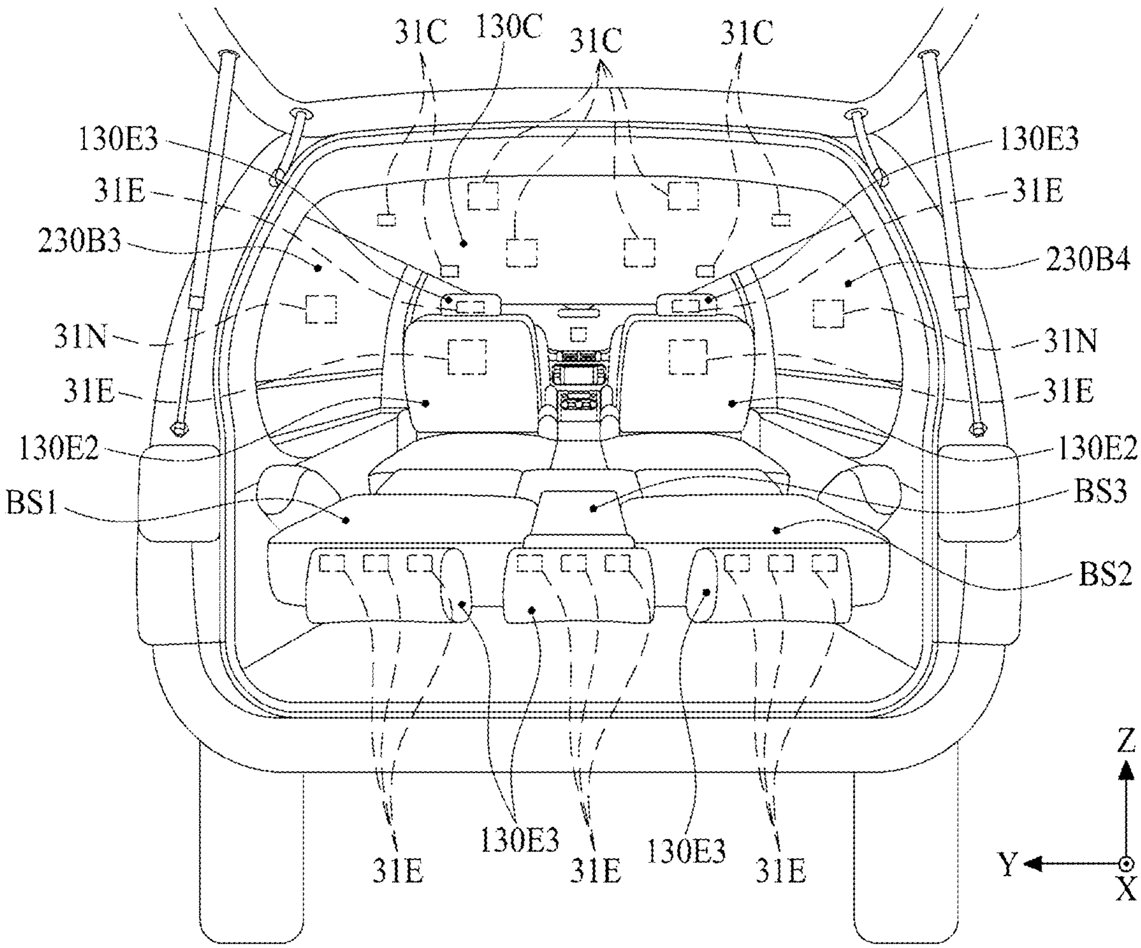

130L2
31L
31C
31C
31C
130C
130I
31I
31C
130L1
31L
230B2(230B)
31C
31C
31C
130: 130C, 130I, 130L
130L: 130L1, 130L2
30-1: 31C
30-2: 31I, 31L
230: 230B

FIG. 13



130: 130C, 130E, 130N 130E: 130E2, 130E3 30-1: 31C, 31E

30-3: 31N 230B: 230B3, 230B4 230: 230B

VIBRATION APPARATUS AND VEHICULAR APPARATUS COMPRISING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of the Korean Patent Application No. 10-2024-0025988 filed on Feb. 22, 2024, which is hereby incorporated by reference as if fully set forth herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a vibration apparatus and vehicular apparatus comprising the same.

Description of the Related Art

[0003] Recently, a demand for a sound apparatus or a sound bar including a vibration apparatus for outputting sound through one or more speakers is increasing.

BRIEF SUMMARY

[0004] However, the sound bar in the related art faces a problem where directionality, sound characteristics, and/or sound pressure characteristics are compromised due to sound interference among the plurality of speakers. The inventors of the present disclosure have recognized the problems and disadvantages of the related art and have performed extensive research and experiments for enhancing a directivity, a sound characteristic and/or a sound pressure level characteristic between multiple speakers. Based on the extensive research and experiments, the inventors have invented a vibration apparatus and a vehicular apparatus including the same, which can enhance a directivity, a sound characteristic and/or a sound pressure level characteristic.

[0005] One or more aspects of the present disclosure are directed to providing a vibration apparatus and a vehicular apparatus including the same, wherein acoustic interference between multiple speakers can be prevented or minimized.

[0006] One or more aspects of the present disclosure are directed to providing a vibration apparatus and a vehicular apparatus including the same, which can improve the directionality, the sound characteristics, and/or the sound pressure characteristics.

[0007] Additional features and advantages of the disclosure will be set forth in the description which follows and in part will be apparent from the description, or may be learned by practice of the disclosure. Other advantages of the present disclosure will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

[0008] To achieve these and other advantages and embodiment of the present disclosure, as embodied and broadly described herein, in one or more embodiments, a vibration apparatus comprises a cover member having first and second areas, a vibration part in each of the first and second areas, and a support member between the cover member and the vibration part.

[0009] To achieve these and other advantages and embodiment of the present disclosure, as embodied and broadly described herein, in one or more embodiments, a vehicular apparatus comprises an interior material exposed to an indoor space, and at least one sound generating apparatus configured to output sound to the indoor space. The at least one sound generating apparatus includes a vibration apparatus comprising a cover member having first and second areas, a vibration part in each of the first and second areas, and a support member between the cover member and the vibration part.

[0010] According to one or more aspects of the present disclosure, the vibration apparatus and the vehicular apparatus comprising the same may prevent or minimize the sound interference between the plurality of speakers.

[0011] According to one or more aspects of the present disclosure, the vibration apparatus and the vehicular apparatus comprising the same may improve the directionality, the sound characteristics, and/or the sound pressure characteristics.

[0012] According to one or more aspects of the present disclosure, the vibration apparatus and the vehicular apparatus comprising the same may improve the sound characteristics and/or the sound pressure characteristics so that it may be driven with low power, thereby reducing power consumption.

[0013] According to one or more aspects of the present disclosure, the vibration apparatus and the vehicular apparatus comprising the same may simplify the structure and may be easily designed.

[0014] An example vehicular apparatus according to one embodiment of the present disclosure includes: a body having a motor mounted therein; a plurality of vehicle interior materials coupled to the body; and a vibration apparatus coupled to at least one vehicle interior material of the plurality of vehicle interior materials.

[0015] The vibration apparatus includes: a cover member having first and second areas; a vibration part in each of the first and second areas; and a support member between the cover member and the vibration part.

[0016] The vibration part includes: a first vibration part in the first area; and a plurality of second vibration parts in the second area. Here, each of the plurality of second vibration parts is smaller than the first vibration part in size.

[0017] Additionally, each of the first vibration part and the plurality of second vibration parts includes: a vibration layer including a piezoelectric material; a first electrode layer on the first surface of the vibration layer; and a second electrode layer on the second surface of the vibration layer. The vibration layer has a first surface and a second surface different from the first surface, and the vibration layers of the first vibration part and the second vibration part have different sizes.

[0018] Other systems, methods, features and advantages will be, or will become, apparent to one with skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features and advantages be included within this description, be within the scope of the present disclosure, and be protected by the following claims. Nothing in this section should be taken as a limitation on those claims. Further aspects and advantages are discussed below in conjunction with aspects of the disclosure.

[0019] It is to be understood that both the foregoing description and the following description of the present disclosure are exemplary and explanatory and are intended to provide further explanation of the disclosure as claimed.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0020] The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this disclosure, illustrate aspects and embodiments of the disclosure and together with the description serve to explain principles of the disclosure.

[0021] FIG. **1** is a plan view illustrating a vibration apparatus according to an aspect of the present disclosure.

[0022] FIG. **2** is a cross-sectional view taken along line I-I' of FIG. **1** according to an aspect of the present disclosure.

[0023] FIG. **3** is a plan view illustrating a vibration apparatus according to another aspect of the present disclosure.

[0024] FIG. **4** is a cross-sectional view taken along line II-II' of FIG. **3** according to an aspect of the present disclosure.

[0025] FIG. **5** is a cross-sectional view taken along line II-II' of FIG. **3** according to another aspect of the present disclosure.

[0026] FIG. **6** is a cross-sectional view taken along line II-II' of FIG. **3** according to another aspect of the present disclosure.

[0027] FIGS. **7**A and **7**B are perspective views illustrating a support member shown in FIG. **6**.

[0028] FIG. **8** is a diagram illustrating a sound apparatus according to an aspect of the present disclosure.

[0029] FIG. **9** is a diagram illustrating a sound apparatus according to another aspect of the present disclosure.

[0030] FIG. **10** is a diagram illustrating a vehicular apparatus according to an aspect of the present disclosure.

[0031] FIG. **11** is a diagram illustrating a vehicular apparatus according to an aspect of the present disclosure.

[0032] FIG. **12** is a diagram illustrating a sound generating apparatus disposed in the roof of the vehicular apparatus of FIGS. **10** and **11**.

[0033] FIG. **13** is a diagram illustrating a sound generating apparatus disposed in the roof and seat of the vehicular apparatus of FIGS. **10** and **11**.

DETAILED DESCRIPTION

[0034] Reference is now made in detail to aspects of the present disclosure, examples of which can be illustrated in the accompanying drawings. In the following description, when a detailed description of well-known functions, structures or configurations can unnecessarily obscure aspects of the present disclosure, a detailed description of such known functions or configurations can have been omitted for brevity. Further, repetitive descriptions can be omitted for brevity. The progression of processing steps and/or operations described is a non-limiting example.

[0035] The sequence of steps and/or operations is not limited to that set forth herein and can be changed to occur in an order that is different from an order described herein, with the exception of steps and/or operations necessarily occurring in a particular order. In one or more examples, two operations in succession can be performed substantially concurrently, or the two operations can be performed in a reverse order or in a different order depending on a function or operation involved.

[0036] Unless stated otherwise, like reference numerals can refer to like elements throughout even when they are shown in different drawings. In one or more aspects, identical elements (or elements with identical names) in different drawings can have the same or substantially the same functions and properties unless stated otherwise. Names of the respective elements used in the following explanations are selected only for convenience and can be thus different from those used in actual products.

[0037] Advantages and features of the present disclosure, and implementation methods thereof, are clarified through the aspects described with reference to the accompanying drawings. The present disclosure may, however, be embodied in different forms and should not be construed as limited to the example aspects set forth herein. Rather, these example aspects are examples and are provided so that this disclosure can be thorough and complete to assist those skilled in the art to understand the inventive concepts without limiting the protected scope of the present disclosure.

[0038] Shapes (e.g., sizes, lengths, widths, heights, thicknesses, locations, radii, diameters, and areas), dimensions, ratios, angles, numbers, and the like disclosed herein, including those illustrated in the drawings, are merely examples, and thus, the present disclosure is not limited to the illustrated details. Any implementation described herein as an "example" is not necessarily to be construed as preferred or advantageous over other implementations. It is, however, noted that the relative dimensions of the components illustrated in the drawings are part of the present disclosure.

[0039] Where a term like "comprise," "have," "include," "contain," "constitute," or the like is used with respect to one or more elements, one or more other elements can be added unless a term such as "only" or the like is used. The terms used in the present disclosure are merely used in order to describe example aspects, and are not intended to limit the scope of the present disclosure. The terms of a singular form can include plural forms unless the context clearly indicates otherwise.

[0040] The word "exemplary" is used to mean serving as an example or illustration, unless otherwise specified. Aspects are example aspects. "Embodiments," "examples," "aspects," and the like should not be construed as preferred or advantageous over other implementations. An embodiment, an example, an example aspect, an aspect, or the like can refer to one or more aspects, one or more examples, one or more example embodiments, one or more embodiments, or the like, unless stated otherwise. Further, the term "may" encompasses all the meanings of the term "can."

[0041] In one or more aspects, unless explicitly stated otherwise, element, feature, or corresponding information (e.g., a level, range, dimension, size, or the like) is construed to include an error or tolerance range even where no explicit description of such an error or tolerance range is provided. An error or tolerance range can be caused by various factors (e.g., process factors, internal or external impact, noise, or the like). In interpreting a numerical value, the value is interpreted as including an error range unless explicitly stated otherwise.

[0042] In describing a positional relationship, when the positional relationship between two parts (e.g., layers, films, regions, components, sections, or the like) is described, for example, using "on," "upon," "on top of," "over," "under," "above," "below," "beneath," "near," "close to," "adjacent to," "beside," "next to," "at or on a side of," or the like, one or more other parts can be located between the two parts unless a more limiting term, such as "immediate(ly)," "direct(ly)," or "close(ly)," is used. For example, where a structure is described as being positioned "on," "upon," "on top of," "over," "under," "above," "below," "beneath," "near," "close to," "adjacent to," "beside," "next to," "at or on a side of," or the like another structure, this description should be construed as including a case in which the structures contact each other as well as a case in which one or more additional structures are disposed or interposed therebetween. Furthermore, the terms "front," "rear," "back," "left," "right," "top," "bottom," "downward," "upward," "upper," "lower," "up," "down," "column," "row," "vertical," "horizontal," and the like refer to an arbitrary frame of reference.

[0043] Spatially relative terms, such as "below," "beneath," "lower," "on," "above," "upper" and the like, can be used to describe a correlation between various elements (e.g., layers, films, regions, components, sections, or the like) as shown in the drawings. The spatially relative terms are to be understood as terms including different orientations of the elements in use or in operation in addition to the orientation depicted in the drawings. For example, if the elements shown in the drawings are turned over, elements described as "below" or "beneath" other elements would be oriented "above" other elements. Thus, the term "below," which is an example term, can include all directions of "above" and "below." Likewise, an exemplary term "above" or "on" can include both directions of "above" and "below."

[0044] In describing a temporal relationship, when the temporal order is described as, for example, "after," "subsequent," "next," "before," "preceding," "prior to," or the like, a case that is not consecutive or not sequential can be included and thus one or more other events can occur therebetween, unless a more limiting term, such as "just," "immediate(ly)," or "direct(ly)," is used.

[0045] The terms, such as "below," "lower," "above," "upper" and the like, can be used herein to describe a relationship between element(s) as illustrated in the drawings. It will be understood that the terms are spatially relative and based on the orientation depicted in the drawings.

[0046] It is understood that, although the terms "first," "second," or the like can be used herein to describe various elements (e.g., layers, films, regions, components, sections, or the like), these elements should not be limited by these terms, for example, to any particular order, precedence, or number of elements. These terms are used only to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. Furthermore, the first element, the second element, and the like can be arbitrarily named according to the convenience of those skilled in the art without departing from the scope of the present disclosure. For clarity, the functions or structures of these elements (e.g., the first element, the second element and the like) are not limited by ordinal numbers or the names in front of the elements. Further, a first element can include one or more first elements. Similarly, a second element or the like can include one or more second elements or the like.

[0047] In describing elements of the present disclosure, the terms "first," "second," "A," "B," "(a)," "(b)," or the like can be used. These terms are intended to identify the corresponding element(s) from the other element(s), and these are not used to define the essence, basis, order, or number of the elements.

[0048] For the expression that an element (e.g., layer, film, region, component, section, or the like) is described as "connected," "coupled," "attached," "adhered," or the like to another element, the element can not only be directly connected, coupled, attached, adhered, or the like to another element, but also be indirectly connected, coupled, attached, adhered, or the like to another element with one or more intervening elements disposed or interposed between the elements, unless otherwise specified.

[0049] For the expression that an element (e.g., layer, film, region, component, section, or the like) "overlaps," or the like with another element, the element can not only directly contact, overlap, or the like with another element, but also indirectly overlap, or the like with another element with one or more intervening elements disposed or interposed between the elements, unless otherwise specified.

[0050] The phrase that an element (e.g., layer, film, region, component, section, or the like) is "provided in," "disposed in," or the like in another element can be understood as that at least a portion of the element is provided in, disposed in, or the like in another element, or that the entirety of the element is provided in, disposed in, or the like in another element. The phrase that an element (e.g., layer, film, region, component, section, or the like) "contacts," "overlaps," or the like with another element can be understood as that at least a portion of the element contacts, overlaps, or the like with a least a portion of another element, that the entirety of the element contacts, overlaps, or the like with a least a portion of another element, or that at least a portion of the element contacts, overlaps, or the like with the entirety of another element.

[0051] The terms such as a "line" or "direction" should not be interpreted only based on a geometrical relationship in which the respective lines or directions are parallel or perpendicular to each other. Such terms can mean a wider range of lines or directions within which the components of the present disclosure can operate functionally. For example, the terms "first direction," "second direction," and the like, such as a direction parallel or perpendicular to "X-axis," "Y-axis," or "Z-axis," should not be interpreted only based on a geometrical relationship in which the respective directions are parallel or perpendicular to each other, and can be meant as directions having wider directivities within the range within which the components of the present disclosure can operate functionally.

[0052] The term "at least one" should be understood as including any and all combinations of one or more of the associated listed items. For example, each of the phrases of "at least one of a first item, a second item, or a third item" and "at least one of a first item, a second item, and a third item", can represent (i) a combination of items provided by two or more of the first item, the second item, and the third item or (ii) only one of the first item, the second item, or the third item.

[0053] The expression of a first element, a second elements, "and/or" a third element should be understood to encompass one of the first, second, and third elements, as well as any and all combinations of the first, second and third elements. By way of example, A, B and/or C encompass only A; only B; only C; any of A, B, and C (e.g., A, B, or C); some combinations of A, B, and C (e.g., A and B; A and

C; or B and C); and all of A, B, and C. Furthermore, an expression "A/B" can be understood as A and/or B. For example, an expression "A/B" can refer to only A; only B; A or B; or A and B.

[0054] In one or more aspects, the terms "between" and "among" can be used interchangeably simply for convenience unless stated otherwise. For example, an expression "between a plurality of elements" can be understood as among a plurality of elements. In another example, an expression "among a plurality of elements" can be understood as between a plurality of elements. In one or more examples, the number of elements can be two. In one or more examples, the number of elements can be more than two. Furthermore, when an element (e.g., layer, film, region, component, sections, or the like) is referred to as being "between" at least two elements, the element can be the only element between the at least two elements, or one or more intervening elements can also be present.

[0055] In one or more aspects, the phrases "each other" and "one another" can be used interchangeably simply for convenience unless stated otherwise. For example, an expression "different from each other" can be understood as different from one another. In another example, an expression "different from one another" can be understood as different from each other. In one or more examples, the number of elements involved in the foregoing expression can be two. In one or more examples, the number of elements involved in the foregoing expression can be more than two.

[0056] In one or more aspects, the phrases "one or more among" and "one or more of" can be used interchangeably simply for convenience unless stated otherwise.

[0057] The term "or" means "inclusive or" rather than "exclusive or." For example, unless otherwise stated or clear from the context, the expression that "x uses a or b" means any one of natural inclusive permutations. For example, "a or b" can mean "a," "b," or "a and b." For example, "a, b or c" can mean "a," "b," "c," "a and b," "b and c," "a and c," or "a, b and c."

[0058] Features of various aspects of the present disclosure can be partially or entirely coupled to or combined with each other, can be technically associated with each other, and can be operated, linked, or driven together in various ways. Aspects of the present disclosure can be implemented or carried out independently from each other, or can be implemented or carried out together in a co-dependent or related relationship. In one or more aspects, the components of each apparatus according to various aspects of the present disclosure can be operatively coupled and configured.

[0059] Unless otherwise defined, the terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example aspects belong. It should be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning, for example, consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense, unless expressly defined otherwise herein.

[0060] The terms used herein have been selected as being general in the related technical field; however, there can be other terms depending on the development and/or change of technology, convention, preference of technicians, and so on. Therefore, the terms used herein should not be understood as limiting technical ideas, but should be understood as examples of the terms for describing example aspects.

[0061] Further, in a specific case, a term can be arbitrarily selected by an applicant, and in this case, the detailed meaning thereof is described herein. Therefore, the terms used herein should be understood based on not only the name of the terms, but also the meaning of the terms and the content hereof.

[0062] "X-axis direction", "Y-axis direction" and "Z-axis direction" should not be construed by a geometric relation only of a mutual vertical relation and can have broader directionality within the range that elements of the present disclosure can act functionally.

[0063] In the following description, various example aspects of the present disclosure are described in detail with reference to the accompanying drawings. With respect to reference numerals to elements of each of the drawings, the same elements can be illustrated in other drawings, and like reference numerals can refer to like elements unless stated otherwise. The same or similar elements can be denoted by the same reference numerals even though they are depicted in different drawings.

[0064] In addition, for convenience of description, a scale, dimension, size, and thickness of each of the elements illustrated in the accompanying drawings can be different from an actual scale, dimension, size, and thickness, and thus, aspects of the present disclosure are not limited to a scale, dimension, size, and thickness illustrated in the drawings.

[0065] FIG. **1** is a plan view illustrating a vibration apparatus according to an aspect of the present disclosure. FIG. **2** is a cross-sectional view taken along line I-I' of FIG. **1** according to an aspect of the present disclosure.

[0066] Referring to FIGS. **1** and **2**, a vibration apparatus **500** according to an embodiment of the present disclosure may be configured to vibrate a vibration member **20**. For example, the vibration apparatus **500** may be connected to the vibration member **20**.

[0067] The vibration member **20** can generate a vibration or can output a sound (or a sound wave), based on the displacements (or driving) of the vibration apparatus **500**. For example, the vibration member **20** can be a diaphragm, a vibration plate, a vibration object, a vibration panel, a sound plate, a sound panel, a passive vibration plate, a passive vibration member, a passive vibration panel, a sound output plate, or a sound vibration plate, but embodiments of the present disclosure are not limited thereto.

[0068] The vibration member **20** can include a single nonmetal material or a composite nonmetal material, but embodiments of the present disclosure are not limited thereto. For example, a single nonmetal material or a composite nonmetal material of the vibration member **20** can include one or more of wood, rubber, plastic, carbon, glass, fiber, cloth, paper, mirror, and leather, but embodiments of the present disclosure are not limited thereto. For example, the paper can be cone paper for speakers. For example, the cone paper can be pulp or foamed plastic, or the like, but embodiments of the present disclosure are not limited thereto.

[0069] According to an embodiment of the present disclosure, the vibration member **20** can be configured in a plastic material such as plastic or styrene material, but embodiments of the present disclosure are not limited thereto.

[0070] The plastic material of the vibration member **20** can be polyethylene terephthalate, polycarbonate, polyimide, polypropylene, polyarylate, polyethersulfone, polyethylene naphthalate, polysulfone, cyclo-olefin copolymer, or carbon fiber reinforced plastic CFRP, or the like, but embodiments of the present disclosure are not limited thereto.

[0071] The styrene material of the vibration member **20** can be an ABS material. The ABS material can be acrylonitrile, butadiene, or styrene.

[0072] A vibration member **20** according to an aspect of the present disclosure may be formed of a porous material. For example, the vibration member **20** may include a micro cellular plastics material. For example, the vibration member **20** may be formed of a micro cellular polyethylene terephthalate material MCPET. The vibration member **20** composed of MCPET has low density and excellent elasticity, whereby it has high sound reproduction capability, thereby improving sound quality.

[0073] According to an aspect of the present disclosure, the vibration member **20** may have a larger area than a vibration apparatus **500**. The vibration apparatus **500** may be configured in a first surface (or upper surface) or a second surface (or lower surface) of the vibration member **20**. The vibration apparatus **500** may have a shape corresponding to that of the vibration member **20** or may have a shape which is the same as that of the vibration member **20**, but aspects of the present disclosure are not limited thereto. For example, the vibration apparatus **500** may have a different shape from the vibration member **20**. For example, the vibration apparatus **500** may include a rectangular shape or a square shape, but aspects of the present disclosure are not limited thereto. For example, the vibration apparatus **500** may have a smaller size than the vibration member **20**.

[0074] According to an aspect of the present disclosure, the vibration apparatus **500** may be disposed or configured in the vibration member **20**. For example, the vibration apparatus **500** may be disposed or configured on one or more of the first surface and the second surface of the vibration member **20**.

[0075] The vibration apparatus **500** can include a piezoelectric material having piezoelectric characteristic. The vibration apparatus **500** can vibrate (or displace or drive) based on driving signals (or a vibration driving signal or a voice signal) applied thereto to vibrate (or displace or drive) the vibration member (or a vehicular interior material). For example, the vibration apparatus **500** can vibrate (or displace or drive) by alternately repeating contraction and/or expansion by the piezoelectric effect (or piezoelectric property). For example, the vibration apparatus **500** can vibrate (or displace or drive) in the vertical direction (or thickness direction) Z by alternately repeating contraction and/or expansion by the inverse piezoelectric effect.

[0076] The vibration apparatus **500** may be formed of a ceramic-based piezoelectric material capable of generating a relatively strong vibration or may be formed of a piezoelectric ceramic material having a perovskite-based crystal structure. For example, the vibration apparatus **500** may be a vibration generating element, a vibration film, a vibration generating film, a vibrator, an active vibrator, an active vibration generator, an actuator, an exciter, a film actuator, a film exciter, an ultrasonic actuator, or an active vibration member, but aspects of the present disclosure are not limited thereto.

[0077] The vibration apparatus **500** according to an aspect of the present disclosure may include a vibration part **510**, a support member **540**, a flexible part **520**, a cover member **560**, and a signal cable **570**.

[0078] The vibration part **510** may be configured to vibrate by a piezoelectric effect according to a driving signal. For example, the vibration part **510** may include a piezoelectric vibration portion. The vibration part **510** may include at least one of a piezoelectric inorganic material and a piezoelectric organic material, but aspects of the present disclosure are not limited thereto. For example, the vibration part **510** may be a vibration element, a piezoelectric element, a piezoelectric element portion, a piezoelectric element layer, a piezoelectric structure, a piezoelectric vibration part, or a piezoelectric vibration layer, but aspects of the present disclosure are not limited thereto.

[0079] The vibration part **510** may be configured in each of a first area A1 and a second area A2 of the cover member **560**. The vibration part **510** may include a first vibration part **511** and a second vibration part **512**. The first vibration part **511** may be configured in the first area A1 of the cover member **560**. The second vibration part **512** may be configured in the second area A2 of the cover member **560**. For example, the vibration part **510** may include one first vibration part **511** in the first area A1 of the cover member **560**. For example, the vibration part **510** may include the plurality of second vibration parts **512** in the second area A2 of the cover member **560**. The plurality of second vibration parts **512** may be configured in parallel to a first direction (or X-axis direction) and a second direction (or Y-axis direction) perpendicular to the first direction, but aspects of the present disclosure are not limited thereto. The first vibration part **511** and the second vibration part **512** may be spaced apart from each other by a predetermined distance. For example, each of the first vibration part **511** and the plurality of second vibration parts **512** may be disposed at fixed intervals.

[0080] The first area A1 may be smaller than the second area A2 in size. The first vibration part **511** and the second vibration part **512** may have different sizes. For example, the second vibration part **512** may have a smaller size than the first vibration part **511**. For example, each of the plurality of second vibration parts **512** may have a smaller size than one first vibration part **511**. For example, the first vibration part **511** and the second vibration part **512** may have a square shape, but aspects of the present disclosure are not limited thereto. For example, a length L2 of one side of the second vibration part **512** may be smaller than a length L1 of one side of the first vibration part **511**. For example, the sound characteristics and/or sound pressure characteristics may be improved according as the size of the vibration part **510** increases, and the directionality may be improved according as the size of the vibration part **510** becomes small.

[0081] According to an aspect of the present disclosure, the first vibration part **511** and the second vibration part **512** are configured to have different sizes so that it is possible to prevent or minimize a sound interference between a plurality of speakers and to improve the directionality, the sound characteristics and/or the sound pressure characteristics. The vibration apparatus **500** according to an aspect of the present disclosure may be applied to a directional speaker.

[0082] According to the embodiment of the present disclosure, each of the first vibration part **511** and the plurality of second vibration part **512** can include a vibration layer

511*a* and **512***a*, a first electrode layer **511***b* and **512***b*, and a second electrode layer **511***c* and **512***c*.

[0083] The vibration layer **511***a* and **512***a* may include a piezoelectric material or an electroactive material including a piezoelectric effect. For example, the piezoelectric material may have a characteristic in which a potential difference is generated by dielectric polarization according to a change in the relative positions of positive (+) ions and negative (−) ions when a pressure or torsion phenomenon is applied to the crystal structure by an external force, and thus vibration is generated by an electric field according to a reverse voltage applied thereto. For example, the vibration layer **511***a* and **512***a* may be a piezoelectric layer, a piezoelectric material layer, an electrically active layer, a piezoelectric composite layer, a piezoelectric composite, or a piezoelectric ceramic composite, but aspects of the present disclosure are not limited thereto.

[0084] The vibration layers **511***a* and **512***a* may be made of a ceramic-based material capable of realizing a relatively strong vibration or a piezoelectric ceramic material having a perovskite-based crystal structure. The perovskite-based crystal structure may have a piezoelectric and/or reverse piezoelectric effect and may have a plate-shaped structure having orientation. The perovskite-based crystal structure may be represented by a chemical formula of ABO_3, wherein the A site may be formed of a divalent metal element, and the B site may be formed of a tetravalent metal element. As an aspect of the present disclosure, in a chemical formula of ABO_3, the A site and the B site may be cations, and O site may be anions. For example, the vibration layers **511***a* and **512***a* may include at least any one of $PbTiO_3$, $PbZrO_3$, $PbZrTiO_3$, $BaTiO_3$, and $SrTiO_3$, but aspects of the present disclosure are not limited thereto.

[0085] The piezoelectric ceramic may be formed of a single crystal ceramic having a single crystal structure, a ceramic material having a polycrystalline structure, or a polycrystalline ceramic. The piezoelectric material of the single crystal ceramic may include α-$AlPO_4$, α-SiO_2, $LiNbO_3$, $Tb_2(MoO_4)_3$, $Li_2B_4O_7$, or ZnO, but aspects of the present disclosure are not limited thereto. The piezoelectric material of the polycrystalline ceramic may include a lead zirconate titanate PZT-based material including lead Pb, zirconium Zr, and titanium Ti, or a lead zirconate nickel niobate PZNN-based material including lead Pb, zirconium Zr, nickel Ni, and niobium Nb, but aspects of the present disclosure are not limited thereto. For example, the vibration layers **511***a* and **512***a* may include at least one selected from the group of materials not including lead Pb, for example, at least one of $CaTiO_3$, $BaTiO_3$, and $SrTiO_3$, but aspects of the present disclosure are not limited thereto.

[0086] Each of the vibration layer **512***a* of the second vibration part **512** and the vibration layer **511***a* of the first vibration part **511** may have a square shape. The vibration layer **512***a* of the second vibration part **512** and the vibration layer **511***a* of the first vibration part **511** may have different sizes. For example, the size of the vibration layer **512***a* of the second vibration part **512** may be smaller than the size of the vibration layer **511***a* of the first vibration part **511**. For example, the length of one side of the vibration layer **512***a* of the second vibration part **512** may be smaller than the length of one side of the vibration layer **511***a* of the first vibration part **511**. For example, the length of one side of the vibration layer **512***a* of the second vibration part **512** may be in a range of 15% to 25% of the length of one side of the vibration layer **511***a* of the first vibration part **511**, but aspects of the present disclosure are not necessarily limited thereto.

[0087] The first electrode layers **511***b* and **512***b* may be disposed on the first surface (or rear surface or lower surface) of the vibration layers **511***a* and **512***a*. The first electrode layers **511***b* and **512***b* may be disposed on the first surface (or rear surface or lower surface) of the vibration layer **511***a* of the first vibration part **511** and the vibration layer **512***a* of the second vibration part **512**. For example, the first electrode layer **511***b* of the first vibration part **511** and the first electrode layer **512***b* of the second vibration part **512** adjacent to each other may be electrically connected to each other. For example, the plurality of first electrode layers **512***b* of the plurality of second vibration parts **512** adjacent to each other may be electrically connected to each other. For example, the first electrode layers **511***b* and **512***b* may be commonly disposed on or connected to the first surface (or lower surface) of the vibration layer **511***a* of the first vibration part **511** and the first surface (or lower surface) of each of the vibration layers **512***a* of the plurality of second vibration parts **512**. For example, the first electrode layers **511***b* and **512***b* may have a cylindrical electrode shape disposed on the entire first surface (or lower surface) of each of the vibration layers **511***a* and **512***a*, but aspects of the present disclosure are not limited thereto.

[0088] The second electrode layers **511***c* and **512***c* may be disposed on the second surface (or front surface or upper surface) of the vibration layers **511***a* and **512***a*. The second electrode layers **511***c* and **512***c* may have the same size as the vibration layers **511***a* and **512***a* or may have a smaller size than the vibration layers **511***a* and **512***a*. For example, the second electrode layers **511***c* and **512***c* may have the same shape as the vibration layers **511***a* and **512***a*, but aspects of the present disclosure are not limited thereto.

[0089] At least one of the first electrode layers **511***b* and **512***b* and the second electrode layers **511***c* and **512***c* according to an aspect of the present disclosure may be formed of a transparent conductive material, a semi-transparent conductive material, or an opaque conductive material. For example, the transparent or semi-transparent conductive material may include indium tin oxide ITO or indium zinc oxide IZO, but aspects of the present disclosure are not limited thereto. The opaque conductive material may include gold Au, silver Ag, platinum Pt, palladium Pd, molybdenum Mo, magnesium Mg, carbon, or silver Ag containing glass frit, or an alloy thereof, but aspects of the present disclosure are not limited thereto. For example, each of the first electrode layers **511***b* and **512***b* and the second electrode layers **511***c* and **512***c* may include silver Ag having low resistivity in order to improve electrical characteristics and/or vibration characteristics of the vibration layers **511***a* and **512***a*. For example, carbon may be a carbon material including carbon black, ketjen black, carbon nanotubes, and graphite, but aspects of the present disclosure are not limited thereto.

[0090] The vibration layers **511***a* and **512***a* may be polarized (or poled) by a constant voltage applied to the first electrode layers **511***b* and **512***b* and the second electrode layers **511***c* and **512***c* under a constant temperature atmosphere or a temperature atmosphere which changes from a high temperature to a room temperature, but aspects of the present disclosure are not limited thereto. For example, the polarization direction (or poling direction) formed in the vibration layers **511**a and **512**a may be formed or aligned (or arranged) from the first electrode layers **511**b and **512**b to the second electrode layers **511**c and **512**c, but aspects of the present disclosure are not limited thereto. For example, the polarization direction (or the poling direction) formed in the vibration layers **511**a and **512**a may be formed or aligned (or arranged) from the second electrode layers **511**c and **512**c to the first electrode layers **511**b and **512**b.

[0091] The vibration layers **511**a and **512**a may vibrate by alternately repeating contraction and/or expansion due to a reverse piezoelectric effect according to a driving signal applied to the first electrode layers **511**b and **512**b and the second electrode layers **511**c and **512**c from the outside. For example, the vibration layers **511**a and **512**a may vibrate in the vertical direction (or thickness direction) and the plan direction by signals applied to the first electrode layers **511**b and **512**b and the second electrode layers **511**c and **512**c. The vibration layers **511**a and **512**a may be displaced (or vibrated or driven) by contraction and/or expansion in the plan direction, thereby improving the vibration characteristics including the sound characteristics and/or the sound pressure characteristics of the vibration apparatus **500**.

[0092] The support member **540** may be configured between the cover member **560** and the vibration part **510**. The support member **540** may be configured between a first cover member **561** and the first vibration part **511**. The support member **540** may be configured between the first cover member **561** and the second vibration part **512**. The thickness of the support member **540** may be 0.5 to 2.0 times the thickness of the vibration layers **511**a and **512**a, but aspects of the present disclosure are not limited thereto. For example, when the thickness of the support member **540** is less than 0.5 times the thickness of the vibration layers **511**a and **512**a, the support member **540** may be damaged by the vibration of the vibration layers **511**a and **512**a. For example, when the thickness of the support member **540** exceeds 2.0 times the thickness of the vibration layers **511**a and **512**a, the vibration of the vibration layers **511**a and **512**a may not be easily transmitted to the vibration member **20**.

[0093] The support member **540** may be configured to support the vibration part **510** and to transmit the vibration of the vibration part **510** to the vibration member **20**. For example, the support member **540** may be connected to the vibration part **510** and may be configured to adjust a resonance frequency of the vibration apparatus **500** to a band of about 40 kHz. For example, according to an aspect of the present disclosure, the resonance frequency of the vibration apparatus **500** may be lowered to the band of about 40 kHz by adjusting the material and thickness of the support member **540**.

[0094] The support member **540** may be connected to the vibration part **510**. The support member **540** may be connected to the first surface (or rear surface) of the first vibration part **511**. The support member **540** may be connected to the first surface (or rear surface) of each of the plurality of second vibration parts **512**. The support member **540** according to an aspect of the present disclosure may include a first support member **541** and a second support member **542**.

[0095] The first support member **541** may transmit the vibration of the first vibration part **511** to the vibration member **20**. The first support member **541** may be configured to adjust the resonance frequency of the first vibration part **511** to the band of about 40 kHz. The first support member **541** may be configured between the cover member **560** and the first vibration part **511**. The first support member **541** may be configured between the first cover member **561** and the first vibration part **511**. The first support member **541** may be connected to the first surface (or rear surface) of the first vibration part **511**.

[0096] The first support member **541** may be configured to convert an in-plane vibration mode (or first vibration mode) of the first vibration part **511** into an out-of-plane vibration mode (or second vibration mode). For example, the first support member **541** may be configured to convert the in-plane vibration mode transmitted to the vibration member **20** into the out-of-plane vibration mode according to the vibration of the first vibration part **511**. For example, the first support member **541** may be configured to convert the in-plane vibration mode according to the vibration of the first vibration part **511** to the out-of-plane vibration mode and to transmit the out-of-plane vibration mode to the vibration member **20**. For example, the first support member **541** may be configured to convert an ultrasonic vibration direction vibration (or transverse vibration) transmitted to the vibration member **20** according to the vibration of the first vibration part **511** into a thickness direction vibration (or longitudinal vibration) of the vibration member **20**. For example, the first support member **541** may be configured to convert the vibration of the first vibration part **511** into a normal direction vibration (or a transverse vibration) with respect to the surface of the first vibration part **511** or a thickness direction vibration (or longitudinal vibration) of the vibration member **20** and to transmit the vibration to the vibration member **20**, thereby increasing (or maximizing) an efficiency of a squeeze film effect generated according to the vibration of the vibration member **20**.

[0097] The second support member **542** may transmit the vibration of the second vibration part **512** to the vibration member **20**. The second support member **542** may be configured to adjust a resonance frequency of the second vibration part **512** to a band of about 40 kHz. The second support member **542** may be configured between the cover member **560** and the second vibration part **512**. The second support member **542** may be configured between the first cover member **561** and the plurality of second vibration parts **512**. The second support member **542** may be connected to each of the first surfaces (or rear surfaces) of the plurality of second vibration parts **512**. For example, the plurality of second support members **542** may be provided, but aspects of the present disclosure are not limited thereto. For example, each of the plurality of second support members **542** may be connected to each of the plurality of second vibration parts **512**.

[0098] The second support member **542** may be configured to convert an in-plane vibration mode (or first vibration mode) of the second vibration part **512** into an out-of-plane vibration mode (or second vibration mode). For example, the second support member **542** may be configured to convert the in-plane vibration mode transmitted to the vibration member **20** into the out-of-plane vibration mode according to the vibration of the second vibration part **512**. For example, the second support member **542** may be configured to convert the in-plane vibration mode according to the vibration of the second vibration part **512** into the out-of-plane vibration mode and to transmit the out-of-plane vibration mode to the vibration member **20**. For example, the second support member **542** may be configured to convert an ultrasonic vibration direction vibration (or transverse vibration) transmitted to the vibration member **20** according to the vibration of the second vibration part **512** into a thickness direction vibration (or longitudinal vibration) of the vibration member **20**. For example, the second support member **542** may convert the vibration of the second vibration part **512** into a normal direction vibration (or transverse vibration) with respect to the surface of the second vibration part **512** or a vibration (or longitudinal vibration) in the thickness direction of the vibration member **20** and to transmit the vibration to the vibration member **20**, thereby increasing (or maximizing) an efficiency of a squeeze film effect generated according to the vibration of the vibration member **20**.

[0099] According to an aspect of the present disclosure, the support member **540** may include at least any one of aluminum Al, copper Cu, and plastic, and the first support member **541** and the second support member **542** may include the same material. For example, the first support member **541** and the second support member **542** may include a metal material or a plastic material. For example, the metal material of the first support member **541** and the second support member **542** may include at least one of stainless steel, aluminum Al, aluminum Al alloy, magnesium Mg, magnesium Mg alloy, copper Cu alloy, and magnesium lithium Mg—Li alloy, but aspects of the present disclosure are not limited thereto. For example, the first support member **541** and the second support member **542** may be formed of a plastic material of a plastic or styrene material, but aspects of the present disclosure are not limited thereto. For example, the plastic material of the first support member **541** and the second support member **542** may be made of polypropylene, polycarbonate, polyethylene terephthalate, polyarylate, polyethylene naphthalate, polysulfone, polyethersulfone, or cyclo-olefin copolymer, but aspects of the present disclosure are not limited thereto. For example, the styrene material may be an ABS material. The ABS material may be Acrylonitrile, Butadiene, and Styrene.

[0100] According to an aspect of the present disclosure, the first support member **541** may have a size corresponding to the first vibration part **511** and the second support member **542** may have a size corresponding to the second vibration part **512**, but aspects of the present disclosure are not limited thereto. For example, the first support member **541** may have a first width W**1** corresponding to the first vibration part **511**. For example, the second support member **542** may have a second width W**2** corresponding to the second vibration part **512**. However, aspects of the present disclosure are not necessarily limited thereto.

[0101] According to an aspect of the present disclosure, the thicknesses of the first support member **541** and the second support member **542** may be different from each other. For example, the thickness T**1** of the first support member **541** may be smaller than the thickness T**2** of the second support member **542**, but aspects of the present disclosure are not limited thereto. For example, the thickness T**1** of the first support member **541** may be 60% or less of the thickness T**2** of the second support member **542**.

[0102] For example, when the sizes of the first vibration part **511** and the second vibration part **512** are different from each other, the resonance frequencies of the ultrasonic waves output from the first vibration part **511** and the second vibration part **512** may be different from each other.

[0103] In the vibration apparatus **500** according to an aspect of the present disclosure, in order to set the resonance frequencies of the ultrasonic waves output from the first vibration part **511** and the second vibration part **512** to be similar to each other (or the range of 40 kHz to 43 kHz), the first support member **541** and the second support member **542** may include the same material and may have different thicknesses, but aspects of the present disclosure are not limited thereto.

[0104] Accordingly, the vibration apparatus **500** according to an aspect of the present disclosure may allow the ultrasonic waves output from the first vibration part **511** and the second vibration part **512** to have similar resonance frequencies. For example, the resonance frequency of the first vibration part **511** according to an aspect of the present disclosure may be 42.9 kHz, and the resonance frequency of the second vibration part **512** may be 42.5 kHz. Accordingly, the vibration apparatus **500** according to an aspect of the present disclosure may improve the sound and/or sound quality characteristics of the vibration apparatus **500** in the first vibration part **511**, and may improve the directionality of the vibration apparatus **500** in the second vibration part **512**.

[0105] The flexible part **520** may be configured to surround each of the first vibration part **511** and the second vibration part **512**. The flexible part **520** may be configured between the first vibration part **511** and the second vibration part **512** adjacent to each other. The flexible part **520** may be configured between each of the plurality of second vibration parts **512**. The flexible part **520** may be configured to surround the remaining surfaces of the vibration part **510** except for the upper and side surfaces of the second electrode layers **511***c* and **512***c*. The flexible part **520** may be configured to surround the side surface of the second support member **542**. The flexible part **520** may be configured to surround the side and rear surfaces of the first support member **541**.

[0106] According to an aspect of the present disclosure, since the vibration apparatus **500** includes the flexible part **520**, vibration energy due to links in the unit lattice of the first vibration part **511** and the second vibration part **512** may be increased, and thus vibration characteristics may be enhanced, and piezoelectric characteristics and flexibility may be secured. For example, the flexible part **520** may be one or more of epoxy-based polymer, acryl-based polymer, and silicone-based polymer, but aspects of the present disclosure are not limited thereto.

[0107] For example, the flexible part **520** may include an organic material portion. For example, the organic material portion may be disposed between each of the vibration layers **511***a* and **512***a* including inorganic material portions so that it is possible to absorb an impact applied to the inorganic material portion (or vibration layer), to improve durability of the vibration part **510** by releasing stress concentrated on the inorganic material portion (or vibration layer), and to provide flexibility to the vibration part **510**.

[0108] For example, the flexible part **520** may include an organic material, an organic polymer, an organic piezoelectric material, or an organic non-piezoelectric material having flexible characteristics compared to the vibration layers **511***a* and **512***a*, but aspects of the present disclosure are not limited thereto. For example, the flexible part **520** may be expressed as an adhesive portion having flexibility, an elastic portion, a bending portion, or a damping portion, but aspects of the present disclosure are not limited thereto.

[0109] According to an aspect of the present disclosure, the vibration apparatus **500** may be in the form of film, may vibrate in the up and down direction by the vibration part **510** having vibration characteristics, and may be bent in a curved shape by the flexible part **520** having flexibility. Also, in the vibration apparatus **500** according to an aspect of the present disclosure, the size of the flexible part **520** may be set according to the piezoelectric characteristics and flexibility required for the vibration apparatus **500**. For example, in the case of the vibration apparatus **500** requiring piezoelectric characteristics rather than flexibility, the size of the flexible part **520** may be reduced. For example, in the case of the vibration apparatus **500** requiring flexibility rather than piezoelectric characteristics, the size of the flexible part **520** may be increased. For example, when the vibration apparatus **500** is attached to the curved vibration member **20**, the shape of the flexible part **520** may be designed according to the shape of the vibration member **20** and the arrangement of the vibration part **510**. Accordingly, the vibration apparatus **500** according to an aspect of the present disclosure may adjust the size and shape of the flexible part **520** according to the size of the vibration apparatus **500** and the shape of the vibration member **20**, thereby facilitating design.

[0110] The vibration apparatus **500** according to an aspect of the present disclosure may include the cover member **560**. The cover member **560** may include the first area A**1** and the second area A**2** adjacent to the first area A**1**. The cover member **560** may include the first cover member **561** and the second cover member **562**.

[0111] The first cover member **561** may be disposed on the first surface (or lower surface) of the vibration part **510**. The first vibration part **511** may be configured in the first area A**1** of the first cover member **561**, and the second vibration part **512** may be configured in the second area A**2** of the first cover member **561**. The first cover member **561** may be disposed on the first electrode layer **511***b* of the first vibration part **511** and the first electrode layer **512***b* of the second vibration part **512**. For example, the first cover member **561** may be configured to cover the first electrode layer **511***b* of the first vibration part **511** and the first electrode layer **512***b* of the second vibration part **512**. The support member **540** may be disposed between the first cover member **561** and the vibration part **510**. For example, the first support member **541** may be configured between the first area A**1** of the first cover member **561** and the first vibration part **511**. For example, the second support member **542** may be configured in the second area A**2** of the first cover member **561** and the second vibration part **512**. The first cover member **561** may be disposed on the first support member **541** and the second support member **542**. For example, the first cover member **561** may be configured to cover the first support member **541** and the second support member **542**. For example, the first cover member **561** may be configured to have a larger size than each of the vibration part **510** and the support member **540**, but aspects of the present disclosure are not limited thereto. The first cover member **561** may be configured to protect the first surface (or lower surface) of the vibration part **510** and the first electrode layers **511***b* and **512***b*. The first cover member **561** may be configured to protect the support member **540** connected to the vibration part **510**.

[0112] The second cover member **562** may be disposed on the second surface (or upper surface) of the vibration part **510**. The second cover member **562** may be configured in the second electrode layers **511***c* and **512***c* of the vibration part **510**. For example, the second cover member **562** may be configured to cover the second electrode layers **511***c* and **512***c* of the vibration part **510**. For example, the second cover member **562** may be in the second electrode layers **511***c* and **512***c*. For example, the second cover member **562** may be configured to have a larger size than the vibration part **510** and may be configured to have the same size as the first cover member **561**. The second cover member **562** may be configured to protect the second surface (or upper surface) of the vibration part **510** and the second electrode layers **511***c* and **512***c*.

[0113] The first cover member **561** and the second cover member **562** according to an aspect of the present disclosure may include the same or different materials. For example, each of the first cover member **561** and the second cover member **562** may be a polyimide film, a polyethylene terephthalate film, or a polyethylene naphthalate film, but aspects of the present disclosure are not limited thereto.

[0114] The first cover member **561** may be connected or coupled to the first surface of the vibration part **510** and/or the first surface of the flexible part **520** through the use of first adhesive layer **563**. For example, the first cover member **561** may be connected or coupled to the first surface of the vibration part **510**, the support member **540**, and/or the first surface of the flexible part **520** by a film laminating process using the first adhesive layer **563**.

[0115] The second cover member **562** may be connected or coupled to the second surface or the second electrode layers **511***c* and **512***c* of the vibration part **510** through the use of second adhesive layer **564**. For example, the second cover member **562** may be connected or coupled to the second surface or the second electrode layers **511***c* and **512***c* of the vibration part **510** by a film laminating process using the second adhesive layer **564**.

[0116] According to an aspect of the present disclosure, each of the first adhesive layer **563** and the second adhesive layer **564** may include an electrically insulating material enabling compression and restoration while having adhesiveness. For example, each of the first adhesive layer **563** and the second adhesive layer **564** may include epoxy resin, acrylic resin, silicone resin, or urethane resin, but aspects of the present disclosure are not limited thereto.

[0117] The first adhesive layer **563** and the second adhesive layer **564** may be configured between the first cover member **561** and the second cover member **562** to surround the vibration part **510**, the support member **540**, and the flexible part **520**. For example, at least one of the first adhesive layer **563** and the second adhesive layer **564** may be configured to surround the vibration part **510**, the support member **540**, and the flexible part **520**.

[0118] The vibration apparatus **500** according to an aspect of the present disclosure may include the signal cable **570**. The signal cable **570** may include a first power supply line (or first signal line) **571** disposed on the first cover member **561** and a second power supply line (or second signal line) **572** disposed on the second cover member **562**.

[0119] The first power supply line **571** may be configured between the vibration part **510** and the first cover member **561**. The first power supply line **571** may be configured between the first electrode layers **511***b* and **512***b* of the vibration part **510** and the first adhesive layer **563**. The first power supply line **571** may be electrically connected to the first electrode layers **511***b* and **512***b* of the vibration part **510**. For example, the first power supply line **571** may be electrically connected to the first electrode layers **511***b* and **512***b* of the vibration part **510** via an anisotropic conductive film. For example, the first power supply line **571** may be electrically connected to the first electrode layers **511***b* and **512***b* of the vibration part **510** through a conductive material (or particles).

[0120] The second power supply line **572** may be configured between the vibration part **510** and the second cover member **562**. The second power supply line **572** may be configured between each of the second electrode layer **511***c* of the first vibration part **511** and the second electrode layer **512***c* of the second vibration part **512** and the second adhesive layer **564**. The second power supply line **572** may be electrically connected to the second electrode layer **511***c* of the first vibration part **511**. The second power supply line **572** may be electrically connected to each of the second electrode layers **512***c* in the plurality of second vibration parts **512**. For example, the second power supply line **572** may be electrically connected to each of the second electrode layers **511***c* and **512***c* of the vibration part **510** via an anisotropic conductive film. For example, the second power supply line **572** may be electrically connected to each of the second electrode layers **511***c* and **512***c* of the vibration part **510** through a conductive material (or particle).

[0121] For example, the signal cable **570** may be electrically connected to a pad portion disposed in the vibration apparatus **500**, and may supply a vibration driving signal (or sound signal) provided from a sound processing circuit to the vibration part **510**. For example, the pad portion may include a flexible printed circuit cable, a flexible flat cable, a single-sided flexible printed circuit, a single-sided flexible printed circuit board, a flexible multilayer printed circuit, or a flexible multilayer printed circuit board, but aspects of the present disclosure are not limited thereto.

[0122] FIG. **3** is a plan view illustrating a vibration apparatus according to another aspect of the present disclosure. FIG. **4** is a cross-sectional view taken along line II-II' of FIG. **3** according to another aspect of the present disclosure. This is substantially the same as the aspect of the present disclosure described with reference to FIGS. **1** and **2**, except that a third area, a third vibration part, and a third support member are additionally provided in the cover member, the vibration part, and the support member, respectively. Thus, only different configurations will be described below.

[0123] Referring to FIGS. **3** and **4**, a vibration part **510** according to another aspect of the present disclosure may be configured in each of a first area **A1** to a third area **A3** of a cover member **560**. The vibration part **510** may include a first vibration part **511** to a third vibration part **513**. The first to third vibration parts **511** to **513** may be respectively configured in the first to third areas **A1** to **A3**.

[0124] According to another aspect of the present disclosure, the first to third areas **A1**, **A2**, and **A3** may have the same size, but aspects of the present disclosure are not limited thereto. For example, the vibration part **510** may include one first vibration part **511** in the first area **A1** of the cover member **560**. For example, the vibration part **510** may include the plurality of second vibration parts **512** in the second area **A2** of the cover member **560**. For example, the vibration part **510** may include one third vibration part **513** in the third area **A3** of the cover member **560**.

[0125] The first vibration part **511** according to another aspect of the present disclosure may have substantially the same configuration as the first vibration part **511** according to the aspect of the present disclosure described with reference to FIGS. **1** and **2**. The number of the plurality of second vibration parts **512** according to another aspect of the present disclosure is different from that of the aspect of the present disclosure described with reference to FIGS. **1** and **2**, and the remaining components may have the same configuration as the second vibration part **512** according to the aspect of the present disclosure described with reference to FIGS. **1** and **2**. Therefore, hereinafter, the third vibration part **513** and the configuration related to the third vibration part **513** will be described.

[0126] According to another aspect of the present disclosure, the first vibration part **511** and the third vibration part **513** may be configured with the second vibration part **512** therebetween, and may have the same size, but aspects of the present disclosure are not limited thereto. For example, the third vibration part **513** may have a square shape, but aspects of the present disclosure are not limited thereto. For example, a length of one side of the third vibration part **513** may be the same as a length of one side of the first vibration part **511**.

[0127] The third vibration part **513** and the second vibration part **512** may have different sizes. For example, the second vibration part **512** may have a smaller size than the third vibration part **513**. The vibration layers **511***a* and **513***a* of the first vibration part **511** and the third vibration part **513** may have the same size. The vibration layers **511***a* and **512***a* of the first vibration part **511** and the second vibration part **512** may have different sizes. The vibration layer **511***a* of the first vibration part **511** may be larger than the vibration layer **512***a* of the second vibration part **512** in size. For example, each of the plurality of second vibration parts **512** may have a smaller size than one third vibration part **513**. For example, the third vibration part **513** and the second vibration part **512** may have a square shape, but aspects of the present disclosure are not limited thereto. For example, a length of one side of the second vibration part **512** may be smaller than the length of one side of the third vibration part **513**, but aspects of the present disclosure are not limited thereto. For example, the sound characteristics and/or sound pressure characteristics may be improved according as the size of the vibration part **510** increases, and the directionality may be improved according as the size of the vibration part **510** becomes small.

[0128] According to the aspect of the present disclosure, the vibration apparatus **500** includes the second vibration part **512** having a different size from the first vibration part **511** and the third vibration part **513** having the same size, thereby preventing or minimizing a sound interference between a plurality of speakers and improving the directionality, the sound characteristics, and/or the sound pressure characteristics. The vibration apparatus **500** according to the aspect of the present disclosure may be applied to a directional speaker.

[0129] According to the aspect of the present disclosure, the third vibration part **513** may include a vibration layer **513***a*, a first electrode layer **513***b*, and a second electrode layer **513***c*. Herein, configurations of the vibration layer **513***a*, the first electrode layer **513***b*, and the second electrode layer **513***c* of the third vibration part **513** are substantially the same as those of the first vibration part **511**.

[0130] According to another aspect of the present disclosure, the size of the vibration layer **512***a* of the second vibration part **512** may be smaller than the size of the vibration layer **513***a* of the third vibration part **513**. For example, the length of one side of the vibration layer **512***a* of the second vibration part **512** may be smaller than the length of one side of the vibration layer **513***a* of the third vibration part **513**, but aspects of the present disclosure are not limited thereto. For example, the length of one side of the vibration layer **512***a* of the second vibration part **512** may be 15% to 25% of the length of one side of the vibration layer **513***a* of the third vibration part **513**, but the aspect of the present disclosure is not necessarily limited thereto.

[0131] The first electrode layers **511***b*, **512***b*, and **513***b* may be disposed on a first surface (or rear surface or lower surface) of the vibration layers **511***a*, **512***a*, and **513***a*. For example, the first electrode layer **511***b* of the first vibration part **511** and the first electrode layer **512***b* of the second vibration part **512** adjacent to each other may be electrically connected to each other. For example, the first electrode layer **513***b* of the third vibration part **513** and the first electrode layer **512***b* of the second vibration part **512** adjacent to each other may be electrically connected to each other. For example, the first electrode layers **511***b*, **512***b*, and **513***b* may be commonly disposed or connected to the first surface (or lower surface) of the vibration layer **511***a* of the first vibration part **511**, the first surface (or lower surface) of each of the vibration layers **512***a* of the plurality of second vibration parts **512**, and the first surface (or lower surface) of the vibration layer **513***a* of the third vibration part **513**. For example, the first electrode layers **511***b*, **512***b*, and **513***b* may have a cylindrical electrode shape disposed on the entire first surface (or lower surface) of each of the vibration layers **511***a*, **512***a*, and **513***a*, but aspects of the present disclosure are not limited thereto.

[0132] The second electrode layer **511***c*, **512***c*, and **513***c* may be disposed on a second surface (or front surface or upper surface) of the vibration layer **511***a*, **512***a*, and **513***a*. The second electrode layers **511***c*, **512***c*, and **513***c* may have the same size as the vibration layers **511***a*, **512***a*, and **513***a* or may have a smaller size than the vibration layers **511***a*, **512***a*, and **513***a*. For example, the second electrode layers **511***c*, **512***c*, and **513***c* may have the same shape as the vibration layers **511***a*, **512***a*, and **513***a*, but aspects of the present disclosure are not limited thereto.

[0133] The vibration layer **513***a*, the first electrode layer **513***b*, and the second electrode layer **513***c* of the third vibration part **513** according to the aspect of the present disclosure may include the same material as the vibration layer **511***a*, the first electrode layer **511***b*, and the second electrode layer **511***c* of the first vibration part **511** described with reference to FIGS. **1** and **2** and may have the same characteristics.

[0134] According to another aspect of the present disclosure, a support member **540** may further include a third support member **543** formed between the cover member **560** and the vibration part **510**. The third support member **543** may be configured between a first cover member **561** and the third vibration part **513**. The third support member **543** may be configured to support the vibration part **510** and to transmit the vibration of the vibration part **510** to a vibration member **20**. The third support member **543** may be connected to the vibration part **510**. The third support member **543** may be connected to the first surface (or rear surface) of the third vibration part **513**.

[0135] The third support member **543** may be configured to convert an in-plane vibration mode (or first vibration mode) of the third vibration part **513** into an out-of-plane vibration mode (or second vibration mode). For example, the third support member **543** may be configured to convert the in-plane vibration mode transmitted to the vibration member **20** into the out-of-plane vibration mode according to the vibration of the third vibration part **513**. For example, the third support member **543** may be configured to convert the in-plane vibration mode according to the vibration of the third vibration part **513** into the out-of-plane vibration mode and transmit the out-of-plane vibration mode to the vibration member **20**. For example, the third support member **543** may be configured to convert an ultrasonic vibration direction vibration (or transverse vibration) transmitted to the vibration member **20** according to the vibration of the third vibration part **513** into a thickness direction vibration (or longitudinal vibration) of the vibration member **20**. For example, the third support member **543** may be configured to convert the vibration of the third vibration part **513** into a normal direction vibration (or transverse vibration) with respect to the surface of the third vibration part **513** or a thickness direction vibration (or longitudinal vibration) of the vibration member **20** and to transmit the vibration to the vibration member **20**, thereby increasing (or maximizing) an efficiency of a squeeze film effect generated according to the vibration of the vibration member **20**.

[0136] According to the aspect of the present disclosure, the third support member **543** may include the same material as the first support member **541**, but aspects of the present disclosure are not limited thereto.

[0137] According to the aspect of the present disclosure, the third support member **543** may have a size corresponding to the third vibration part **513** and may have a third width W**3** corresponding to the third vibration part **513**, but aspects of the present disclosure are not limited thereto. For example, the third width W**3** may be equal to the first width W**1** and greater than the second width W**2**, but aspects of the present disclosure are not limited thereto.

[0138] According to the aspect of the present disclosure, the thicknesses of the third support member **543** and the first support member **541** may be the same, and the thicknesses of the third support member **543** and the second support member **542** may be different from each other. For example, the thicknesses T**1** and T**3** of each of the first support member **541** and the third support member **543** may be smaller than the thickness T**2** of the second support member **542**, but aspects of the present disclosure are not limited thereto. For example, each of the thicknesses T**1** and T**3** of the first support member **541** and the third support member **543** may be 60% or less of the thickness T**2** of the second support member **542**.

[0139] For example, when the sizes of the first vibration part **511** and the second vibration part **512** are different from each other, the resonance frequencies of the ultrasonic waves output from the first vibration part **511** and the second vibration part **512** may be different from each other. For example, when the sizes of the third vibration part **513** and the second vibration part **512** are different from each other, the resonance frequencies of the ultrasonic waves output from the third vibration part **513** and the second vibration part **512** may be different from each other.

[0140] In the vibration apparatus **500** according to the aspect of the present disclosure, in order to set the resonance frequencies of the ultrasonic waves output from the first vibration part **511** to the third vibration part **513** to be similar to each other (or the range of 40 kHz to 43 kHz), the first support member **541** to the third support member **543** may include the same material, and the second support member **542** having the different thickness may be disposed between the first support member **541** and the third support member **543**.

[0141] Accordingly, the vibration apparatus **500** according to the aspect of the present disclosure may allow the ultrasonic waves output from the first vibration part **511** to the third vibration part **513** to have similar resonance frequencies. For example, the resonance frequency of the first vibration part **511** and the third vibration part **513** according to the aspect of the present disclosure may be 42.9 kHz, and the resonance frequency of the second vibration part **512** may be 42.5 kHz. Accordingly, the vibration apparatus **500** according to the aspect of the present disclosure may improve the sound and/or sound quality characteristics of the vibration apparatus **500** in the first vibration part **511** and the third vibration part **513** and may improve the directionality of the vibration apparatus **500** in the second vibration part **512**. For example, the sound of low-pitched sound band may be output from the first vibration part **511** and the third vibration part **513**, and the sound of high-pitched sound band may be output from the second vibration part **512**.

[0142] The flexible part **520** may be configured to surround each of the first vibration part **511** to the third vibration part **513**. The flexible part **520** may be configured between the second vibration part **512** and the third vibration part **513** adjacent to each other. The flexible part **520** may be configured to surround the remaining surfaces of the vibration part **510** except for the upper and side surfaces of the second electrode layers **511***c*, **512***c*, and **513***c*. The flexible part **520** may be configured to surround the side surface and the rear surface of the third support member **543**.

[0143] According to the aspect of the present disclosure, since the vibration apparatus **500** includes the flexible part **520**, vibration energy due to links in the unit lattice of the first vibration part **511** to the third vibration part **513** may be increased, and thus vibration characteristics may be enhanced, and piezoelectric characteristics and flexibility may be secured.

[0144] According to the aspect of the present disclosure, the cover member **560** may include the first area A**1** to the third area A**3**. The first to third areas A**1** to A**3** may be configured in parallel. The first area A**1** and the third area A**3** may be spaced apart from each other with the second area A**2** therebetween. The first to third areas A**1** to A**3** may have the same size, but aspects of the present disclosure are not limited thereto.

[0145] The first cover member **561** may include the first area A**1** to the third area A**3**. The first cover member **561** may include the first area A**1** to the third area A**3** which are parallel to each other. The third vibration part **513** may be configured in the third area A**3** of the first cover member **561**. The first cover member **561** may be formed on the first electrode layer **511***b* of the first vibration part **511**, the first electrode layer **512***b* of the second vibration part **512**, and the first electrode layer **513***b* of the third vibration part **513**. For example, the first cover member **561** may be configured to cover the first electrode layer **511***b* of the first vibration part **511**, the first electrode layer **512***b* of the second vibration part **512**, and the first electrode layer **513***b* of the third vibration part **513**. For example, the third support member **543** may be configured between the third area A**3** of the first cover member **561** and the third vibration part **513**. The first cover member **561** may be configured in the third support member **543**. For example, the first cover member **561** may be configured to cover the third support member **543**. The first cover member **561** may be configured to protect the first surface (or lower surface) of the vibration part **510** and the first electrode layers **511***b*, **512***b*, and **513***b*. The first cover member **561** may be configured to protect the first to third support members **541**, **542**, and **543**.

[0146] The second cover member **562** may be configured to cover the second electrode layers **511***c*, **512***c*, and **513***c* of the vibration part **510**. For example, the second cover member **562** may be in the second electrode layers **511***c*, **512***c*, and **513***c*. The second cover member **562** may be configured to protect the second surface (or upper surface) of the vibration part **510** and the second electrode layers **511***c*, **512***c*, and **513***c*.

[0147] The second cover member **562** may be connected or coupled to the second surface of the vibration part **510** or the second electrode layers **511***c*, **512***c*, and **513***c* via the second adhesive layer **564**. For example, the second cover member **562** may be connected or coupled to the second surface of the vibration part **510** or the second electrode layers **511***c*, **512***c*, and **513***c* by a film laminating process using the second adhesive layer **564**.

[0148] A first power supply line **571** according to another aspect of the present disclosure may be configured between the first electrode layers **511***b*, **512***b*, and **513***b* of the vibration part **510** and the first adhesive layer **563**. The first power supply line **571** may be electrically connected to the first electrode layers **511***b*, **512***b*, and **513***b* of the vibration part **510**. For example, the first power supply line **571** may be electrically connected to the first electrode layers **511***b*, **512***b*, and **513***b* of the vibration part **510** via an anisotropic conductive film. For example, the first power supply line **571** may be electrically connected to the first electrode layers **511***b*, **512***b*, and **513***b* of the vibration part **510** through a conductive material (or particles).

[0149] A second power supply line **572** according to another aspect of the present disclosure may be configured between each of the second electrode layer **511***c* of the first vibration part **511**, the second electrode layer **512***c* of the second vibration part **512**, and the second electrode layer **513***c* of the third vibration part **513** and the second adhesive layer **564**. The second power supply line **572** may be electrically connected to the second electrode layer **513***c* of the third vibration part **513**.

[0150] FIG. **5** is a cross-sectional view taken along line II-II' of FIG. **3** according to another aspect of the present disclosure. In FIG. **5**, the configuration of the support member is changed, and the remaining configuration other than the support member may be substantially the same as the other aspect described with reference to FIGS. **3** and **4**. Accordingly, different configurations will be described below.

[0151] Referring to FIGS. **3** and **5**, the support member **540** according to another aspect of the present disclosure may include first to third support members **541**, **542**, and **543** formed between a cover member **560** and a vibration part **510**.

[0152] According to another aspect of the present disclosure, the first support member **541** and the third support member **543** may include the same material, but aspects of the present disclosure are not limited thereto. The first support member **541** and the second support member **542** may include different materials, but aspects of the present disclosure are not limited thereto. The third support member **543** and the second support member **542** may include different materials, but aspects of the present disclosure are not limited thereto. For example, the first support member **541** and the third support member **543** may include copper Cu, and the second support member **542** may include aluminum Al, but aspects of the present disclosure are not limited thereto. For example, the first to third support members **541**, **542** and **543** may have the same thickness T**1**, T**2**, and T**3**, but aspects of the present disclosure are not limited thereto.

[0153] For example, when the sizes of the first vibration part **511** and the second vibration part **512** are different from each other, the resonance frequencies of the ultrasonic waves output from the first vibration part **511** and the second vibration part **512** may be different from each other. For example, when the sizes of the third vibration part **513** and the second vibration part **512** are different from each other, the resonance frequencies of the ultrasonic waves output from the third vibration part **513** and the second vibration part **512** may be different from each other.

[0154] In the vibration apparatus **500** according to another aspect of the present disclosure, in order to set the resonance frequencies of the ultrasonic waves output from the first vibration part **511** to the third vibration part **513** to be similar to each other (or the range of 40 kHz to 43 kHz), the first support member **541** and the third support member **543** may be made with the material different from that of the second support member **542**, and the second support member **542** of the different material may be configured between the first support member **541** and the third support member **543**.

[0155] Accordingly, the vibration apparatus **500** according to another aspect of the present disclosure may allow the ultrasonic waves output from the first vibration part **511** to the third vibration part **513** to have similar resonance frequencies. For example, according to another aspect of the present disclosure, the resonance frequency of the first vibration part **511** and the third vibration part **513** may be 42.9 kHz, and the resonance frequency of the second vibration part **512** may be 42.5 kHz. Accordingly, the vibration apparatus **500** according to the aspect of the present disclosure may improve the sound quality characteristics of the vibration apparatus **500** in the first vibration part **511** and the third vibration part **513** and may improve the directionality of the vibration apparatus **500** in the second vibration part **512**. The vibration apparatus **500** according to the aspect of the present disclosure may be applied to a directional speaker.

[0156] According to another aspect of the present disclosure, since the thicknesses of the first support member **541** to the third support member **543** are the same, the flexible part **520** may be configured to surround the side surfaces of the first support member **541** to the third support member **543**.

[0157] FIG. **6** is a cross-sectional view taken along line II-II' of FIG. **3** according to another aspect of the present disclosure. FIGS. **7**A and **7**B are perspective views illustrating a support member shown in FIG. **6**. This may be substantially the same as the other aspects of the present disclosure described with reference to FIGS. **3** and **4**, except that the configuration of the support member is changed. Thus, different configurations will be described as follows.

[0158] Referring to FIGS. **3**, **6**, **7**A, and **7**B, the support member **540** according to another aspect of the present disclosure may further include first to third support members **541**, **542**, and **543** configured between a cover member **560** and a vibration part **510**. For example, the first to third support members **541**, **542**, and **543** may include the same material, but aspects of the present disclosure are not limited thereto.

[0159] According to another aspect of the present disclosure, the first to third support members **541**, **542**, and **543** may include support plates **541***a*, **542***a*, and **543***a* and protrusions **541***b*, **542***b*, and **543***b*, respectively.

[0160] According to the aspect of the present disclosure, each of the support plates **541***a*, **542***a*, and **543***a* may support each of first to third vibration parts **511**, **512**, and **513**. The respective support plates **541***a*, **542***a*, and **543***a* may have the same size as the first to third vibration parts **511**, **512**, and **513**, but aspects of the present disclosure are not limited thereto. For example, the thicknesses of the support plates **541***a* and **543***a* respectively provided in the first support member **541** and the third support member **543** may be the same, but aspects of the present disclosure are not limited thereto. For example, the thicknesses of the support plates **541***a* and **542***a* respectively provided in the first support member **541** and the second support member **542** may be different from each other, but aspects of the present disclosure are not limited thereto. For example, the support plate **541***a* of the first support member **541** and the support plate **542***a* of the second support member **542** may have different thicknesses, but aspects of the present disclosure are not limited thereto. For example, the thickness P**1** of the support plate **541***a* of the first support member **541** may be smaller than the thickness P**2** of the support plate **542***a* of the second support member **542**, but aspects of the present disclosure are not limited thereto. For example, the thicknesses of the support plates **543***a* and **542***a* in the third support member **543** and the second support member **542** may be different from each other, but aspects of the present disclosure are not limited thereto. For example, the support plate **543***a* of the third support member **543** and the support plate **542***a* of the second support member **542** may have different thicknesses, but aspects of the present disclosure are not limited thereto. For example, the thickness P**3** of the support plate **543***a* of the third support member **543** may be smaller than the thickness P**2** of the support plate **542***a* of the second support member **542**, but aspects of the present disclosure are not limited thereto.

[0161] According to the aspect of the present disclosure, each of the protrusions **541***b*, **542***b*, and **543***b* may vertically protrude from each of the support plates **541***a*, **542***a*, and **543***a*. For example, each of the protrusions **541***b*, **542***b*, and **543***b* may support each of the support plates **541***a*, **542***a*, and **543***a*. Each of the support plates **541***a*, **542***a*, and **543***a* may be configured with a pair of protrusions **541***b*, **542***b*, and **543***b* vertically configured from facing sides. For example, the thicknesses PT**1** and PT**3** of the protrusions **541***b* and

543*b* in the first support member **541** and the third support member **543** may be the same, but aspects of the present disclosure are not limited thereto. For example, the thicknesses PT**1** and PT**2** of the protrusions **541***b* and **542***b* in the first support member **541** and the second support member **542** may be different from each other, but aspects of the present disclosure are not limited thereto. For example, the thicknesses PT**1** and PT**2** of the protrusion **541***b* in the first support member **541** and the protrusion **542***b* in the second support member **542** may be different from each other, but aspects of the present disclosure are not limited thereto. For example, the thickness PT**1** of the protrusion **541***b* of the first support member **541** may be greater than the thickness PT**2** of the protrusion **542***b* of the second support member **542**, but aspects of the present disclosure are not limited thereto. For example, the thicknesses PT**3** and PT**2** of the protrusions **543***b* and **542***b* in the third support member **543** and the second support member **542** may be different from each other, but aspects of the present disclosure are not limited thereto. For example, the protrusions **543***b* of the third support member **543** and the protrusions **542***b* of the second support member **542** may have different thicknesses PT**3** and PT**2**, but aspects of the present disclosure are not limited thereto. For example, the thickness PT**3** of the protrusion **543***b* of the third support member **543** may be greater than the thickness PT**2** of the protrusion **542***b* of the second support member **542**, but aspects of the present disclosure are not limited thereto.

[0162] For example, when the sizes of the first vibration part **511** and the second vibration part **512** are different from each other, the resonance frequencies of the ultrasonic waves output from the first vibration part **511** and the second vibration part **512** may be different from each other. For example, when the sizes of the third vibration part **513** and the second vibration part **512** are different from each other, the resonance frequencies of the ultrasonic waves output from the third vibration part **513** and the second vibration part **512** may be different from each other.

[0163] In the vibration apparatus **500** according to the aspect of the present disclosure, in order to set the resonance frequencies of the ultrasonic waves output from the first vibration part **511** to the third vibration part **513** to be similar to each other (or the range of 40 kHz to 43 kHz), the first support member **541** to the third support member **543** may include the same material, and the second support member **542** may be disposed between the first support member **541** and the third support member **543**.

[0164] Also, the vibration apparatus **500** according to another aspect of the present disclosure may be configured such that the first to third support members **541**, **542**, and **543** includes the support plates **541***a*, **542***a*, and **543***a* and the protrusions **541***b*, **542***b*, and **543***b*, and the thicknesses of the support plates **541***a*, **542***a*, and **543***a* and the protrusions **541***b*, **542***b*, and **543***b* in the respective support members **541**, **542**, **543** may be adjusted.

[0165] Accordingly, the vibration apparatus **500** according to the aspect of the present disclosure may allow the ultrasonic waves output from the first vibration part **511** to the third vibration part **513** to have similar resonance frequencies. For example, the resonance frequency of the first vibration part **511** and the third vibration part **513** according to the aspect of the present disclosure may be 42.9 kHz, and the resonance frequency of the second vibration part **512** may be 42.5 kHz. Accordingly, the vibration apparatus **500** according to the aspect of the present disclosure may improve the sound characteristics and/or sound quality characteristics of the vibration apparatus **500** in the first vibration part **511** and the third vibration part **513** and may improve the directionality of the vibration apparatus **500** in the second vibration part **512**. The vibration apparatus **500** according to the aspect of the present disclosure may be applied to a directional speaker.

[0166] According to another aspect of the present disclosure, the flexible part **520** may be configured to surround each of the first vibration part **511** to the third vibration part **513**. The flexible part **520** may be between the first vibration part **511** and the second vibration part **512** adjacent to each other and between the second vibration part **512** and the third vibration part **513**. The flexible part **520** may be disposed on a rear surface of the support plate **541***a* and **543***a* in each of the first support member **541** and the third support member **543**. The flexible part **520** may be configured to surround the rear surface of each of the first vibration part **511** to the third vibration part **513**. The flexible part **520** may be filled between the rear surface of the support plate **541***a* of the first support member **541** and the facing protrusion **541***b*. The flexible part **520** may be filled between the rear surface of the support plate **542***a* of the second support member **542** and the facing protrusion **542***b*. The flexible part **520** may be filled between the rear surface of the support plate **543***a* of the third support member **543** and the facing protrusion **543***b*.

[0167] According to another aspect of the present disclosure, the first cover member **561** may be configured to protect the first surface (or lower surface) of the vibration part **510** and the first electrode layers **511***b*, **512***b*, and **513***b*. The first cover member **561** may be configured to protect the first to third support members **541**, **542**, and **543**. The first cover member **561** may be configured in the first to third support members **541**, **542**, and **543**. The first cover member **561** may cover the first to third support members **541**, **542**, and **543**.

[0168] FIG. **8** is a diagram illustrating a sound apparatus according to an aspect of the present disclosure. FIG. **8** illustrates a vehicular sound apparatus according to an embodiment of the present disclosure.

[0169] Referring to FIG. **8**, a sound apparatus **30** according to an embodiment of the present disclosure can be disposed or equipped at an inner portion of a vehicle **10** to output a sound S toward an indoor space IS of the vehicle **10**.

[0170] The vehicle **10** can be a vehicular apparatus including one or more seats and one or more windows. For example, the vehicle **10** can include an automobile, a train, a ship, or an aircraft, but the embodiments of the present disclosure are not limited thereto.

[0171] The vehicle **10** according to an embodiment of the present disclosure can include a main structure **110**.

[0172] The main structure **110** of the vehicle **10** can include a main frame, a sub-frame, a side frame, a door frame, an under frame, a seat frame, etc., but the embodiments of the present disclosure are not limited thereto. For example, the main structure **110** can be a body, a vehicle structure, a frame structure, etc., but the embodiments of the present disclosure are not limited thereto.

[0173] The vehicle **10** according to the embodiment of the present disclosure can further include an exterior material **120**.

[0174] The exterior material **120** of the vehicle **10** can be configured to cover the main structure **110**. For example, the exterior material **120** can be configured to cover the exterior of the main structure **110**. In the following description, the exterior material **120** can be a vehicle exterior material **120**, and these may be used interchangeably.

[0175] The exterior material **120** of the vehicle **10** can include a hood panel, a front fender panel, a dash panel, a filler panel, a trunk panel, a roof panel (or a ceiling), a floor panel, a door inner panel, a door outer panel, and the like, but the embodiments of the present disclosure are not limited thereto.

[0176] The exterior material **120** according to the embodiment of the present disclosure can include at least one of a flat portion and a curved portion (or a bent portion). For example, the exterior material **120** can have a structure corresponding to the structure of the corresponding main structure **110** or can have a structure different from the structure of the corresponding main structure **110**.

[0177] The vehicle **10** according to the embodiment of the present disclosure can further include an interior material **130**. In the following description, the interior material **130** can be referred to as a vehicle interior material **130**, and these may be used interchangeably.

[0178] The vehicle interior material **130** can include all elements configuring an inner portion of the vehicle **10**, or can include all elements disposed at the indoor space IS of the vehicle **10**. For example, the vehicle interior material **130** can be an interior member or an inner finish material of the vehicle **10**, but embodiments of the present disclosure are not limited thereto.

[0179] The vehicle interior material **130** according to the embodiment of the present disclosure can be configured to cover at least one of the main structure **110** and the exterior material **120** in the indoor space IS of the vehicle **10**. For example, the interior material **130** can be configured to be exposed to the indoor space IS of the vehicle **10** while covering at least one of the main structure **110** and the exterior material **120** in the indoor space IS of the vehicle **10**.

[0180] The vehicle interior material **130** according to an embodiment of the present disclosure can be exposed at the inner portion or the indoor space IS of the vehicle **10** in the inner portion or the indoor space IS of the vehicle **10**. For example, the vehicle interior material **130** can be configured to cover one surface (or an interior surface) of at least one or more of a main frame (or a vehicle body), a side frame (or a side body), a door frame (or a door body), a handle frame (or a steering hub), and a seat frame, which are exposed at the indoor space IS of the vehicle **10**.

[0181] The vehicle interior material **130** according to an embodiment of the present disclosure can include a dashboard, a pillar interior material (or a pillar trim), a floor interior material (or a floor carpet), a roof interior material (or a headliner), a door interior material (or a door trim), a handle interior material (or a steering cover), a seat interior material, a rear package interior material (or a back seat shelf), an overhead console (or an indoor illumination interior material), a rear view mirror, a glove box, and a sun visor, or the like, but embodiments of the present disclosure are not limited thereto.

[0182] The vehicle interior material **130** according to an embodiment of the present disclosure can include one or more materials of metal, wood, rubber, plastic, glass, fiber, cloth, paper, a mirror, leather, and carbon, but embodiments of the present disclosure are not limited thereto. The vehicle interior material **130** including a plastic material can be an injection material which is implemented by an injection process using a thermoplastic resin or a thermosetting resin, but embodiments of the present disclosure are not limited thereto. The vehicle interior material **130** including a fiber material can include at least one or more of a plastic composite fiber, a carbon fiber (or an aramid fiber), and a natural fiber, but embodiments of the present disclosure are not limited thereto. The vehicle interior material **130** including the fiber material can include a textile sheet, a knit sheet, or a nonwoven fabric, or the like, but embodiments of the present disclosure are not limited thereto. For example, the paper can be cone paper for speakers. For example, the cone paper can be pulp or foamed plastic, or the like, but embodiments of the present disclosure are not limited thereto. The vehicle interior material **130** including a leather material can include natural leather or artificial leather, but embodiments of the present disclosure are not limited thereto.

[0183] The vehicle interior material **130** according to an embodiment of the present disclosure can include at least one or more of a flat portion (or flat surface portion) and a curved portion (or curved surface portion). For example, the vehicle interior material **130** can have a surface structure corresponding to a surface structure of a corresponding vehicle structure, or can have a surface structure which differs from a surface structure of a corresponding vehicle structure.

[0184] According to an embodiment of the present disclosure, the sound apparatus **30** can be disposed at the vehicle interior material **130**. The sound apparatus **30** can vibrate the vehicle interior material **130** to generate a sound S based on a vibration of the vehicle interior material **130**. For example, the sound apparatus **30** can directly vibrate the vehicle interior material **130** to generate the sound S based on a vibration of the vehicle interior material **130**. For example, the sound apparatus **30** can be configured to vibrate the vehicle interior material **130** to output the sound S toward an inner portion or an indoor space IS of the vehicle **10**. Thus, the sound apparatus **30** can use the vehicle interior material **130** as a sound vibration plate or an acoustic diaphragm. The vehicle interior material **130** can perform a function of a vibration plate, a sound vibration plate, or a sound generating plate for outputting the sound S. For example, the vehicle interior material **130** can have a size which is greater than the sound apparatus **30**, but embodiments of the present disclosure are not limited thereto.

[0185] According to an embodiment of the present disclosure, the sound apparatus **30** can be disposed in at least one or more of a dashboard, a pillar interior material, a floor interior material, a roof interior material, a door interior material, a handle interior material, and a seat interior material, or can be positioned or connected (or coupled) to at least one of a rear package interior material, an overhead console, a rear view mirror, a glove box, and a sun visor. For example, the sound apparatus **30** can be configured to vibrate (or directly vibrate) at least one or more of a dashboard, a pillar interior material, a floor interior material, a roof interior material, a door interior material, a handle interior material, a seat interior material, a rear package interior material, an overhead console, a rear view mirror, a glove box, and a sun visor to output sound S toward the interior and/or indoor space IS of the vehicle **10**.

[0186] A vehicle **10** according to an embodiment of the present disclosure can include one or more sound apparatus **30** arranged at or connected to at least one area (or part) of a vehicle interior material **130**. The one or more sound apparatus **30** can vibrate at least one area (or part) of the vehicle interior material **130** to output realistic sound S and/or three-dimensional sound including multi-channels toward the indoor space IS of the vehicle **10**.

[0187] According to an embodiment of the present disclosure, the sound apparatus **30** can be configured in an area between the vehicle interior material **130** and the main structure **110** or an area between the vehicle interior material **130** and the exterior material **120**. The sound apparatus **30** is placed in an area between the vehicle interior material **130** and the main structure **110** or an area between the vehicle interior material **130** and the exterior material **120**, and can output sound S by indirectly or directly vibrating at least one of the areas between the vehicle interior material **130** and the main structure **110** and the areas between the vehicle interior material **130** and the exterior material **120**.

[0188] According to an embodiment of the present disclosure, the sound apparatus **30** can be arranged in one or more of the region (or first region) between the main structure **110** and the exterior material **120**, the region (or second region) between the main structure **110** and the vehicle interior material **130**, the exterior material **120**, and the vehicle interior material **130**. For example, the sound apparatus **30** can be arranged in one or more of the region (or first region) between the main structure **110** and the exterior material **120**, the region (or second region) between the main structure **110** and the vehicle interior material **130**, the exterior material **120**, and the vehicle interior material **130**, and can be configured to output sound. For example, the sound apparatus **30** is placed in one or more of the area between the main structure **110** and the exterior material **120** (or the first area), the area between the main structure **110** and the vehicle interior material **130** (or the second area), the exterior material **120**, and the vehicle interior material **130**, and can output sound by indirectly or directly vibrating one or more of the main structure **110**, the exterior material **120**, and the vehicle interior material **130**.

[0189] According to an embodiment of the present disclosure, the sound apparatus **30** can be configured to output sound between the exterior material **120** and the interior material **130** of the vehicle **10**. For example, the sound apparatus **30** can be disposed in at least one of a region (or a third region) between the exterior material **120** and the interior material **130**, the exterior material **120**, and the interior material **130**. For example, the sound apparatus **30** can be disposed between the exterior material **120** and the interior material **130**, and can indirectly or directly vibrate at least one of the exterior material **120** and the interior material **130** to output sound. For example, the sound apparatus **30** is connected or coupled between the exterior material **120** and the vehicle interior material **130** to at least one of the exterior material **120** and the vehicle interior material **130**, and can output sound by indirectly or directly vibrating at least one of the exterior material **120** and the vehicle interior material **130**. For example, at least one of the exterior material **120** and the vehicle interior material **130** can output sound S according to the driving (or vibration or displacement) of at least one sound apparatus **30**.

[0190] According to an embodiment of the present disclosure, at least one of the exterior material **120** and the interior material **130** of the vehicle **10** can be a vibration plate, an acoustic vibration plate, or a sound generating plate for outputting sound S. For example, since each of the exterior material **120** and the vehicle interior material **130** for outputting sound S has a larger size (or area) than the sound apparatus **30**, it can function as a large-area vibration plate, acoustic vibration plate, or sound generating plate, thereby improving acoustic characteristics and/or sound pressure characteristics of a sound range including a low-pitched sound range generated by the sound apparatus **30**. For example, the sound frequency of the low-pitched sound range can be 300 Hz or less, 400 Hz or less, or 500 Hz or less, but the embodiments of the present disclosure are not limited thereto.

[0191] FIG. **9** is a diagram illustrating a sound apparatus according to another aspect of the present disclosure. FIG. **9** illustrates an embodiment implemented by modifying the vehicle interior material **130** on which the sound apparatus described above with reference to FIG. **8** is disposed. Therefore, in the following description, repetitive descriptions of elements other than a vehicle interior material **130** are omitted or will be briefly given.

[0192] Referring to FIG. **9**, the sound apparatus **30** according to another embodiment of the present disclosure can vibrate a vehicle interior material **130** to output sound S.

[0193] The vehicle interior material **130** according to anther embodiment of the present disclosure can include one or more materials of metal, wood, rubber, plastic, glass, fiber, cloth, paper, a mirror, a leather and carbon, but embodiments of the present disclosure are not limited thereto.

[0194] The vehicle interior material **130** according to another embodiment of the present disclosure can include a base member **131** and a surface member **133**. For example, the base member **131** can be an injection material, a first interior material, an inner interior material, or a rear interior material, but embodiments of the present disclosure are not limited thereto. The surface member **133** can be a second interior material, an outer interior material, a front interior material, an outer surface member, a reinforcement member, or a decoration member, but embodiments of the present disclosure are not limited thereto.

[0195] The base member **131** can include a plastic material, but embodiments of the present disclosure are not limited thereto. The base member **131** according to an embodiment of the present disclosure can include an injection material. For example, the base member **131** can be an injection material which is implemented by an injection process using a thermoplastic resin or a thermosetting resin, but embodiments of the present disclosure are not limited thereto. The vehicle interior material **130** or the base member **131** can be configured to cover an inner portion of a vehicle **10**. For example, the vehicle interior material **130** or the base member **131** can be configured to cover at least one of the main structure **110** and the exterior material **120** in the indoor space IS of the vehicle **10**. For example, the vehicle interior material **130** or the base member **131** can be configured to cover one surface (or an inner surface) of at least one or more of a main frame, a side frame, a door frame, and a handle frame, which are exposed at the indoor space IS of the vehicle **10**.

[0196] The base member **131** can include one or more of a flat portion and a curved portion. For example, the base member **131** can have a surface structure corresponding to a surface structure of a corresponding vehicle structure, or can have a surface structure which differs from a surface structure of a corresponding vehicle structure.

[0197] The surface member **133** can be disposed on the base member **131**. The surface member **133** can cover the base member **131** at the inner portion or the indoor space IS of the vehicle **10** and can be exposed at the inner portion or the indoor space IS of the vehicle **10**. For example, the surface member **133** can be disposed at or coupled to a front surface (or an interior surface) of the base member **131** exposed at the indoor space IS of the vehicle **10**.

[0198] The vehicle interior material **130** or the surface member **133** according to an embodiment of the present disclosure can include one or more materials of a fiber, leather, cloth, and wood, but embodiments of the present disclosure are not limited thereto. For example, the surface member **133** can be a fiber material. For example, the surface member **133** including a fiber material can include one or more of a synthetic fiber, a carbon fiber (or an aramid fiber), and a natural fiber, but embodiments of the present disclosure are not limited thereto. For example, the surface member **133** including the fiber material can be a textile sheet, a knit sheet, or a nonwoven fabric, but embodiments of the present disclosure are not limited thereto. For example, the surface member **133** including the fiber material can be a fabric member, but embodiments of the present disclosure are not limited thereto.

[0199] The synthetic fiber can be a thermoplastic resin and can include a polyolefin-based fiber which is an eco-friendly material which does not relatively release a harmful substance. For example, the polyolefin-based fiber can include a polyethylene fiber, a polypropylene fiber, or a polyethylene terephthalate fiber, but embodiments of the present disclosure are not limited thereto. The polyolefin-based fiber can be a fiber of a single resin or a fiber of a core-shell structure, but embodiments of the present disclosure are not limited thereto. The natural fiber can be a composite fiber of one or two or more of a jute fiber, a kenaf fiber, an abaca fiber, a coconut fiber, and a wood fiber, but embodiments of the present disclosure are not limited thereto.

[0200] The sound apparatus **30** can be covered by the vehicle interior material **130**. The sound apparatus **30** can be configured to vibrate the vehicle interior material **130** including the base member **131** and the surface member **133** to output a sound S toward the inner portion or the indoor space IS of the vehicle **10**.

[0201] The sound apparatus **30** according to another embodiment of the present disclosure can output the sound S toward the indoor space IS of the vehicle **10** by using the vehicle interior material as a vibration plate or a sound vibration plate and can output a realistic sound S or stereo sound including a multichannel toward the indoor space IS of the vehicle **10**.

[0202] According to another embodiment of the present disclosure, the sound apparatus **30** can be surrounded by an enclosure. A space can be provided between the sound apparatus **30** and the enclosure, and a sound (or a sound pressure level) can be output based on a vibration of the vehicle interior material **130** based on air between the sound apparatus **30** and the enclosure. However, there can be a problem where a sound of the low—pitched sound band is not output, and there can be a problem where the flatness of a sound pressure level is reduced because the number of peaks and dips occurring in a reproduction frequency band of a sound (or a sound pressure level) generated based on a vibration of the vehicle interior material **130** increases and each of a highest sound pressure level and a lowest sound pressure level occurring in a reproduction frequency band of a sound (or a sound pressure level) generated based on a vibration of the vehicle interior material **130** increases. The peak can be a phenomenon where a sound pressure level bounces in a specific frequency, and the dip can be a phenomenon where a low sound pressure level is generated as the occurrence of a sound having a specific frequency is reduced. The flatness of a sound characteristic can be a level of a deviation between a highest sound pressure and a lowest sound pressure.

[0203] FIG. **10** is a diagram illustrating a vehicular apparatus according to an aspect of the present disclosure. FIG. **11** is a diagram illustrating a vehicular apparatus according to an aspect of the present disclosure. FIG. **12** is a diagram illustrating a sound generating apparatus disposed in the roof of the vehicular apparatus of FIGS. **10** and **11**. FIG. **13** is a diagram illustrating a sound generating apparatus disposed in the roof and seat of the vehicular apparatus of FIGS. **10** and **11**.

[0204] Referring to FIGS. **10** to **13**, a vehicular apparatus **10** according to an embodiment of the present disclosure can include a vehicle interior material **130** and a first sound generating apparatus **30-1**. For example, the vehicle interior material **130** can output a sound based on a vibration of one or more first sound generating apparatuses **30-1**. For example, the vehicle interior material **130** can be exposed at an indoor space.

[0205] The first sound generating apparatus **30-1** can include at least one or more of sound generating modules **31**A to **31**G which are disposed at one or more of a dashboard **130**A, a pillar interior material **130**B, a roof interior material **130**C, a door interior material **130**D, a seat interior material **130**E, a handle interior material **130**F, and a floor interior material **130**G. For example, the first sound generating apparatus **30-1** can include at least one or more of the first to seventh sound generating modules **31**A to **31**G, and thus, can output sounds of one or more channels.

[0206] The first sound generating module **31**A according to an embodiment of the present disclosure can be disposed between the dashboard **130**A and a dash panel and can vibrate the dashboard **130**A to output a sound based on a vibration of the dashboard **130**A. According to an embodiment of the present disclosure, the first sound generating module **31**A can include a sound apparatus **30** described above with reference to FIGS. **1** to **7**, and thus, repeated descriptions are omitted. For example, the first sound generating module **31**A can be a dashboard speaker.

[0207] According to an embodiment of the present disclosure, at least one or more of the dash panel and the dashboard **130**A can include a first region corresponding to a driver seat DS, a second region corresponding to a passenger seat PS, and a third region (or a middle region) between the first region and the second region. At least one or more of the dash panel and the dashboard **130**A can include a fourth region which is inclined to face the passenger seat PS. According to an embodiment of the present disclosure, the first sound generating module **31**A can be disposed to vibrate at least one or more of the first to fourth regions of the dashboard **130**A. For example, the first sound generating module **31**A can be disposed at each of the first and second regions of the dashboard **130**A, or can be disposed at each of the first to fourth regions of the dashboard **130**A. For example, the first sound generating module **31**A can be disposed at each of the first and second regions of the dashboard **130**A, or can be disposed at at least one or more of the first to fourth regions of the dashboard **130**A. For example, the first sound generating module **31**A can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the first sound generating module **31**A configured to vibrate at least one or more of the first to fourth regions of the dashboard **130**A can have a same sound output characteristic or different sound output characteristics. For example, the first sound generating module **31**A configured to vibrate each of the first to fourth regions of the dashboard **130**A can have a same sound output characteristic or different sound output characteristics.

[0208] The second sound generating module **31**B according to an embodiment of the present disclosure can be disposed between the pillar interior material **130**B and a pillar panel and can vibrate the pillar interior material **130**B to output a sound based on a vibration of the pillar interior material **130**B. For example, the second sound generating module **31**B can directly vibrate the pillar interior material **130**B to output a sound based on a vibration of the pillar interior material **130**B. For example, the second sound generating module **31**B can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the second sound generating module **31**B can be a pillar speaker or a tweeter speaker.

[0209] According to an embodiment of the present disclosure, the pillar panel can include a first pillar (or an A pillar) disposed at both sides of a front glass window, a second pillar (or a B pillar) disposed at both sides of a center of a vehicle body, and a third pillar (or a C pillar) disposed at both sides of a rear portion of the vehicle body. The pillar interior material **130**B can include a first pillar interior material **130**B**1** covering the first pillar, a second pillar interior material **130**B**2** covering the second pillar, and a third pillar interior material **130**B**3** covering the third pillar. According to an embodiment of the present disclosure, the second sound generating module **31**B can be disposed at at least one or more of a region between the first pillar and the first pillar interior material **130**B**1**, a region between the second pillar and the second pillar interior material **130**B**2**, and a region between the third pillar and the third pillar interior material **130**B**3**, and thus, can vibrate at least one or more of the first to third pillar interior materials **130**B**1** to **130**B**3**. For example, the second sound generating module **31**B can be configured to output a sound at about 2 kHz to about 20 kHz, but embodiments of the present disclosure are not limited thereto. For example, the second sound generating module **31**B can be configured to output a sound at about 150 Hz to about 20 kHz. For example, the second sound generating module **31**B configured to vibrate at least one or more of the first to third pillar interior materials **130**B**1** to **130**B**3** can have a same sound output characteristic or different sound output characteristics. The various sound generating modules can be provided in a vehicle display, such as a speedometer or a passenger display such as a detachable tablet or other displays.

[0210] Referring to FIGS. **10** to **12**, the third sound generating module **31**C according to an embodiment of the present disclosure can be disposed between the roof interior material **130**C and a roof frame (or a roof panel) and can vibrate the roof interior material **130**C to output a sound based on a vibration of the roof interior material **130**C. For example, the third sound generating module **31**C can directly vibrate the roof interior material **130**C to output a sound based on a vibration of the roof interior material **130**C. For example, the third sound generating module **31**C can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the third sound generating module **31**C can be a roof speaker.

[0211] Referring to FIGS. **10** to **12**, the roof interior material **130**C can include the first region corresponding to the driver seat DS, the second region corresponding to the passenger seat PS, a third region corresponding to a region between the driver seat DS and the passenger seat PS, a fourth region corresponding to a first rear seat BS**1** behind the driver seat DS, a fifth region corresponding to a second rear seat BS**2** behind the passenger seat PS, a sixth region corresponding to a third rear seat BS**3** between the first rear seat BS**1** and the second rear seat BS**2**, and a seventh region between the third region and the sixth region. For example, the third sound generating module **31**C can be disposed to vibrate at least one or more among the first to seventh regions of the roof interior material **130**C. For example, the third sound generating module **31**C can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the third sound generating module **31**C configured to vibrate at least one or more of the first to seventh regions of the roof interior material **130**C can have a same sound output characteristic or different sound output characteristics. For example, at least one or more of third sound generating modules **31**C configured to vibrate each of the first to seventh regions of the roof interior material **130**C can be configured to output a sound of about 2 kHz to about 20 kHz, and the other third sound generating modules **31**C can be configured to output a sound of about 150 Hz to about 20 kHz.

[0212] Referring to FIGS. **10** and **11**, the fourth sound generating module **31**D according to an embodiment of the present disclosure can be disposed between the door frame and the door interior material **130**D and can vibrate the door interior material **130**D to output a sound based on a vibration of the door interior material **130**D. According to an embodiment of the present disclosure, the fourth sound generating module **31**D can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the fourth sound generating module **31**D can be a door speaker.

[0213] According to an embodiment of the present disclosure, at least one or more of the door frame and the door interior material **130**D can include an upper region, a middle region, and a lower region with respect to a height direction Z of the vehicular apparatus **10**. For example, the fourth sound generating module **31**D can be disposed at at least one or more of the upper region, the middle region, and the lower region between the door frame and the door interior material **130**D, and thus, can vibrate at least one or more of the upper region, the middle region, and the lower region of the door interior material **130**D.

[0214] According to an embodiment of the present disclosure, the upper region of the door interior material **130**D can have a relatively small curvature radius. The fourth sound generating module **31**D for vibrating the upper region of the door interior material **130**D can include the sound apparatus having a flexible characteristic.

[0215] According to an embodiment of the present disclosure, the door frame can include a first door frame (or a left front door frame), a second door frame (or a right front door frame), a third door frame (or a left rear door frame), and a fourth door frame (or a right rear door frame). According to an embodiment of the present disclosure, the door interior material **130**D can include a first door interior material (or a left front door interior material) **130**D**1** covering the first door frame, a second door interior material (or a right front door interior material) **130**D**2** covering the second door frame, a third door interior material (or a left rear door interior material) **130**D**3** covering the third door frame, and a fourth door interior material (or a right rear door interior material) **130**D**4** covering the fourth door frame. For example, the fourth sound generating module **31**D can be disposed at at least one or more of an upper region, a middle region, and a lower region between each of the first to fourth door frames and the first to fourth door interior materials **130**D**1** to **130**D**4** and can vibrate at least one or more of the upper region, the middle region, and the lower region of each of the first to fourth door interior materials **130**D**1** to **130**D**4**.

[0216] According to an embodiment of the present disclosure, the fourth sound generating module **31**D configured to vibrate the upper region of each of the first to fourth door interior materials **130**D**1** to **130**D**4** can be configured to output a sound of about 2 kHz to about 20 kHz, or can be configured to output a sound of about 150 Hz to about 20 kHz.

[0217] According to an embodiment of the present disclosure, the fourth sound generating module **31**D configured to vibrate the middle regions and the lower regions of at least one or more of the first to fourth door interior materials **130**D**1** to **130**D**4** can be configured to output a sound of about 150 Hz to about 20 kHz. The fourth sound generating module **31**D configured to vibrate the middle region and the lower region of each of the first to fourth door interior materials **130**D**1** to **130**D**4** can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the fourth sound generating module **31**D configured to vibrate the middle regions and the lower regions of at least one or more of the first to fourth door interior materials **130**D**1** to **130**D**4** can be one or more of a woofer, a mid-woofer, and a sub-woofer. For example, the fourth sound generating module **31**D configured to vibrate the middle region and the lower region of each of the first to fourth door interior materials **130**D**1** to **130**D**4** can be one or more of a woofer, a mid-woofer, and a sub-woofer.

[0218] Sounds, which are respectively output from the fourth sound generating module **31**D disposed at the first door interior material **130**D**1** and the fourth sound generating module **31**D disposed at the second door interior material **130**D**2**, can be combined and output. For example, sounds, which are respectively output from at least one or more of the fourth sound generating module **31**D disposed at the first door interior material **130**D**1** and the fourth sound generating module **31**D disposed at the second door interior material **130**D**2**, can be combined and output. In addition, sounds, which are respectively output from the fourth sound generating module **31**D disposed at the third door interior material **130**D**3** and the fourth sound generating module **31**D disposed at the fourth door interior material **130**D**4** can be combined and output.

[0219] According to an embodiment of the present disclosure, the upper region of each of the first to fourth door interior materials **130**D**1** to **130**D**4** can include a first upper region adjacent to the dashboard **130**A, a second upper region adjacent to the rear seats BS**1** to BS**3**, and a third upper region between the first upper region and the second upper region. For example, the fourth sound generating module **31**D can be disposed at one or more of the first to third upper regions of each of the first to fourth door interior materials **130**D**1** to **130**D**4**. For example, the fourth sound generating module **31**D can be disposed at the first upper region of each of the first and second door interior materials **130**D**1** and **130**D**2** and can be disposed at one or more of the second and third upper regions of each of the first and second door interior materials **130**D**1** and **130**D**2**. For example, the fourth sound generating module **31**D can be disposed at one or more among the first to third upper regions of one or more of the first to fourth door interior materials **130**D**1** to **130**D**4**. For example, the fourth sound generating module **31**D configured to vibrate the first upper regions of one or more of the first and second door interior materials **130**D**1** and **130**D**2** can be configured to output a sound of about 2 kHz to about 20 kHz, and the fourth sound generating module **31**D configured to vibrate one or more among the second and third upper regions of each of the first and second door interior materials **130**D**1** and **130**D**2** can be configured to output a sound of about 2 kHz to about 20 kHz, or can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the fourth sound generating module **31**D configured to vibrate one or more among the second and third upper regions of at least one or more of the first and second door interior materials **130**D**1** and **130**D**2** can be configured to output a sound of about 2 kHz to about 20 kHz, or can be configured to output a sound of about 150 Hz to about 20 kHz.

[0220] Referring to FIGS. **10**, **11**, and **13**, the fifth sound generating module **31**E according to an embodiment of the present disclosure can be disposed between a seat frame and the seat interior material **130**E and can vibrate the seat interior material **130**E to output a sound based on a vibration of the seat interior material **130**E. For example, the fifth sound generating module **31**E can directly vibrate the seat interior material **130**E to output a sound based on a vibration of the seat interior material **130**E. For example, the fifth sound generating module **31**E can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the fifth sound generating module **31**E can be a sheet speaker or a headrest speaker.

[0221] According to an embodiment of the present disclosure, the seat frame can include a first seat frame (or a driver seat frame), a second seat frame (or a passenger seat frame), a third seat frame (or a first rear seat frame), a fourth seat frame (or a second rear seat frame), and a fifth seat frame (or a third rear seat frame). According to an embodiment of the present disclosure, the seat interior material **130**E can include the first seat interior material surrounding the first seat frame, the second seat interior material surrounding the second seat frame, the third seat interior material surrounding the third seat frame, the fourth seat interior material surrounding the fourth seat frame, and the fifth seat interior material surrounding the fifth seat frame.

[0222] According to an embodiment of the present disclosure, at least one or more of the first to fifth seat frames can include a seat bottom frame, a seat back frame, and a headrest frame. The seat interior material **130**E can include a seat bottom interior material **130**E**1** surrounding the seat bottom frame, a seat back interior material **130**E**2** surrounding the seat back frame, and a headrest interior material **130**E**3** surrounding the headrest frame. At least one or more of the seat bottom interior material **130**E**1**, the seat back interior material **130**E**2**, and the headrest interior material **130**E**3** can include an inner interior material and an outer interior material. For example, the inner interior material can include a foam layer. For example, the outer interior material can include a surface layer including a fiber or leather. For example, the outer interior material can further include a base layer including a plastic material which supports the surface layer.

[0223] According to an embodiment of the present disclosure, the fifth sound generating module **31**E can be disposed at at least one or more of a region between the seat back frame and the seat back interior material **130**E**2** and a region between the headrest frame and the headrest interior material **130**E**3**, and thus, can vibrate at least one or more of the outer interior material of the seat back interior material **130**E**2** and the outer interior material of the headrest interior material **130**E**3**.

[0224] According to an embodiment of the present disclosure, the fifth sound generating module **31**E disposed at at least one or more of the driver seat DS and the passenger seat PS can be disposed at at least one or more of the region between the seat back frame and the seat back interior material **130**E**2** and the region between the headrest frame and the headrest interior material **130**E**3**.

[0225] According to an embodiment of the present disclosure, the fifth sound generating module **31**E disposed at at least one or more of the first to third rear seats BS**1** to BS**3** can be disposed between the headrest frame and the headrest interior material **130**E**3**. For example, at least one or more of the first to third rear seats BS**1** to BS**3** can include at least one or more fifth sound generating modules **31**E disposed between the headrest frame and the headrest interior material **130**E**3**.

[0226] According to an embodiment of the present disclosure, the fifth sound generating module **31**E vibrating the seat back interior materials **130**E**2** of at least one or more of the driver seat DS and the passenger seat PS can be configured to output a sound of about 150 Hz to about 20 kHz. According to an embodiment of the present disclosure, the fifth sound generating module **31**E vibrating the headrest interior materials **130**E**3** of at least one or more of the driver seat DS, the passenger seat PS, and the first to third rear seats BS**1** to BS**3** can be configured to output a sound of about 2 kHz to about 20 kHz, or can be configured to output a sound of about 150 Hz to about 20 kHz.

[0227] Referring to FIGS. **10** and **11**, the sixth sound generating module **31**F according to an embodiment of the present disclosure can be disposed between a handle frame and the handle interior material **130**F and can vibrate the handle interior material **130**F to output a sound based on a vibration of the handle interior material **130**F. For example, the sixth sound generating module **31**F can directly vibrate the handle interior material **130**F to output a sound based on a vibration of the handle interior material **130**F. According to an embodiment of the present disclosure, the sixth sound generating module **31**F can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the sixth sound generating module **31**F can be a handle speaker or a steering speaker.

[0228] According to an embodiment of the present disclosure, the sixth sound generating module **31**F can directly vibrate the handle interior material **130**F to provide the driver with the sound based on the vibration of the handle interior material **130**F. A sound output by the sixth sound generating module **31**F can be a sound which is a same as or different from a sound output from each of the first to fifth sound generating modules **31**A to **31**E. As an embodiment of the present disclosure, the sixth sound generating module **31**F can output a sound which is to be provided to only the driver. As another embodiment of the present disclosure, the sound output by the sixth sound generating module **31**F and a sound output by each of the first to fifth sound generating modules **31**A to **31**E can be combined and output.

[0229] Referring to FIGS. **10** and **11**, the seventh sound generating module **31**G can be disposed between the floor panel and the floor interior material **130**G and can indirectly or directly vibrate the floor interior material **130**G to output a sound based on a vibration of the floor interior material **130**G.

[0230] The seventh sound generating module **31**G can be disposed between the floor panel and the floor interior material **130**G between the front seats DS and PS and the third rear seat BS**3**. For example, the seventh sound generating module **31**G can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the seventh sound generating module **31**G can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the seventh sound generating module **31**G can be a floor speaker, a bottom speaker, or an under speaker.

[0231] Referring to FIGS. **10** to **12**, the vehicular apparatus or the vehicle **10** according to an embodiment of the present disclosure can further include a second sound generating apparatus **30-2** which is disposed at the vehicle interior material **130** exposed at an indoor space. For example, the vehicular apparatus or the vehicle **10** according to an embodiment of the present disclosure can include only the second sound generating apparatus **30-2** instead of the first sound generating apparatus **30-1**, or can include all of the first sound generating apparatus **30-1** and the second sound generating apparatus **30-2**.

[0232] According to an embodiment of the present disclosure, the vehicle interior material **130** can further include a rear view mirror **130**H, an overhead console **130**I, a rear package interior material **130**J, a glove box **130**K, and a sun visor **130**L, or the like.

[0233] The second sound generating apparatus **30-2** can include at least one or more of sound generating modules **31**H to **31**L which are disposed at at least one of the rear view mirror **130**H, the overhead console **130**I, the rear package interior material **130**J, the glove box **130**K, and the sun visor **130**L. For example, the second sound generating apparatus **30-2** can include at least one or more of eighth to twelfth sound generating modules **31**H to **31**L, and thus, can output sounds of one or more channels.

[0234] Referring to FIGS. **10** to **12**, the eighth sound generating module **31**H can be disposed at the rear view mirror **130**H and can indirectly or directly vibrate the rear view mirror **130**H to output a sound based on a vibration of the rear view mirror **130**H.

[0235] The eighth sound generating module **31**H can be disposed between a mirror housing connected to a vehicle body structure and the rear view mirror **130**H supported by the mirror housing. According to an embodiment of the present disclosure, the eighth sound generating module **31**H can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the eighth sound generating module **31**H can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the eighth sound generating module **31**H can be a mirror speaker.

[0236] Referring to FIGS. **12** and **13**, the ninth sound generating module **31**I can be disposed at the overhead console **130**I and can indirectly or directly vibrate a console cover of the overhead console **130**I to output a sound based on a vibration of an interior material (or the console cover) of the overhead console **130**I.

[0237] According to an embodiment of the present disclosure, the overhead console **130**I can include a console box buried (or embedded) into the roof panel, a lighting device disposed at the console box, and the console cover covering the lighting device and the console box.

[0238] The ninth sound generating module **31**I can be disposed between the console cover and the console box of the overhead console **130**I and can vibrate the console cover. For example, the ninth sound generating module **31**I can be disposed between the console cover and the console box of the overhead console **130**I and can directly vibrate the console cover. According to an embodiment of the present disclosure, the ninth sound generating module **31**I can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the ninth sound generating module **31**I can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the ninth sound generating module **31**I can be a console speaker or a lighting speaker.

[0239] According to an embodiment of the present disclosure, the vehicular apparatus or the vehicle **10** can further include a center lighting box disposed at a center region of the roof interior material **130**C, a center lighting device disposed at the center lighting box, and a center lighting cover covering the center lighting device. In this case, the ninth sound generating module **31**I can be further disposed between a center lighting cover and a center lighting box and can additionally vibrate the center lighting cover.

[0240] Referring to FIGS. **10** and **11**, the tenth sound generating module **31**J can be disposed at the rear package interior material **130**J and can indirectly or directly vibrate the rear package interior material **130**J to output a sound based on a vibration of the rear package interior material **130**J.

[0241] The rear package interior material **130**J can be disposed behind the first to third rear seats BS**1** to BS**3**. For example, a portion of the rear package interior material **130**J can be disposed under a rear glass window **230**C.

[0242] The tenth sound generating module **31**J can be disposed at a rear surface of the rear package interior material **130**J and can indirectly or directly vibrate the rear package interior material **130**J. According to an embodiment of the present disclosure, the tenth sound generating module **31**J can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the tenth sound generating module **31**J can be a rear speaker.

[0243] According to an embodiment of the present disclosure, the rear package interior material **130**J can include a first region corresponding to a rear portion of the first rear seat BS**1**, a second region corresponding to a rear portion of the second rear seat BS**2**, and a third region corresponding to a rear portion of the third rear seat BS**3**. According to an embodiment of the present disclosure, the tenth sound generating module **31**J can be disposed to vibrate at least one or more of the first to third regions of the rear package interior material **130**J. For example, the tenth sound generating module **31**J can be disposed at each of the first and second regions of the rear package interior material **130J**, or can be disposed at each of the first to third regions of the rear package interior material **130**J. For example, the tenth sound generating module **31**J can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the tenth sound generating module **31**J configured to vibrate at least one or more of the first to third regions of the rear package interior material **130**J can have a same sound output characteristic or different sound output characteristics.

[0244] Referring to FIGS. **10** and **11**, the eleventh sound generating module **31**K can be disposed at a glove box **130**K and can indirectly or directly vibrate the glove box **130**K to output a sound based on a vibration of the glove box **130**K.

[0245] The glove box **130**K can be disposed at a dashboard **130**A corresponding to a front portion of the passenger seat PS.

[0246] The eleventh sound generating module **31**K can be disposed at an inner surface of the glove box **130**K and can vibrate the glove box **130**K. For example, the eleventh sound generating module **31**K can directly vibrate the glove box **130**K. According to an embodiment of the present disclosure, the eleventh sound generating module **31**K can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the eleventh sound generating module **31**K can be configured to output a sound of about 150 Hz to about 20 kHz, or can be one or more of a woofer, a mid-woofer, and a sub-woofer. For example, the eleventh sound generating module **31**K can be a glove box speaker.

[0247] Referring to FIG. **12**, the twelfth sound generating module **31**L can be disposed at the sun visor **130**L and can vibrate the sun visor **130**L to output a sound based on a vibration of the sun visor **130**L. For example, the twelfth sound generating module **31**L can directly vibrate the sun visor **130**L to output the sound based on the vibration of the sun visor **130**L.

[0248] The sun visor **130**L can include a first sun visor **130**L**1** corresponding to the driver seat DS and a second sun visor **130**L**2** corresponding to the passenger seat PS.

[0249] The twelfth sound generating module **31**L can be disposed at at least one or more of the first sun visor **130**L**1** and the second sun visor **130**L**2** and can vibrate at least one or more of the first sun visor **130**L**1** and the second sun visor **130**L**2**. For example, the twelfth sound generating module **31**L can directly vibrate at least one or more of the first sun visor **130**L**1** and the second sun visor **130**L**2**. According to an embodiment of the present disclosure, the twelfth sound generating module **31**L can include a sound apparatus **30**

described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted. For example, the twelfth sound generating module **31**L can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the twelfth sound generating module **31**L can be a sun visor speaker.

[0250] According to an embodiment of the present disclosure, at least one or more of the first sun visor **130**L**1** and the second sun visor **130**L**2** can further include a sun visor mirror. In this case, the twelfth sound generating module **31**L can be configured to vibrate a sun visor mirror of at least one or more of the first sun visor **130**L**1** and the second sun visor **130**L**2**. The twelfth sound generating module **31**L can directly vibrate the sun visor mirror of at least one or more of the first sun visor **130**L**1** and the second sun visor **130**L**2**. The twelfth sound generating module **31**L vibrating the sun visor mirror can include a sound apparatus **30** described above with reference to FIGS. **1** to **6**, and thus, repeated descriptions are omitted.

[0251] Referring to FIGS. **10** to **12**, the vehicular apparatus or the vehicle **10** according to an embodiment of the present disclosure can further include a third sound generating apparatus **30-3** disposed at a vehicle glass window **230**. For example, the vehicular apparatus or the vehicle **10** according to an embodiment of the present disclosure can include the third sound generating apparatus **30-3** instead of at least one or more of the first and second sound generating apparatuses **30-1** and **30-2**, or can include all of the first to third sound generating apparatuses **30-1** to **30-3**.

[0252] The third sound generating apparatus **30-3** can include at least one or more sound generating modules **31**N and **31**O disposed at the vehicle glass window **230**. For example, the third sound generating apparatus **30-3** can include at least one or more of thirteenth and fourteenth sound generating modules **31**N and **31**O, and thus, can output sounds of one or more channels. For example, the third sound generating apparatus **30-3** can be a transparent sound generating apparatus.

[0253] The vehicle glass window **230** of the vehicular apparatus or the vehicle **10** can include at least one or more of a front glass window and a side glass window. The vehicle glass window **230** of the vehicular apparatus can further include at least one or more of a rear glass window and a roof glass window.

[0254] The vehicle glass window **230** according to an embodiment of the present disclosure can be configured to be wholly transparent. The vehicle glass window **230** according to another embodiment of the present disclosure can include a transparent portion and a semitransparent portion surrounding the transparent portion. The vehicle glass window **230** according to another embodiment of the present disclosure can include a transparent portion and an opaque portion surrounding the transparent portion.

[0255] The vibration apparatus **500** described above with reference to FIGS. **1** to **6** can be configured to be transparent or semitransparent. For example, when the vehicle glass window **230** is wholly transparent, the vibration apparatus **500** can be configured to be transparent and can be disposed at a middle region or a peripheral region of the vehicle glass window **230**. When the vehicle glass window **230** includes the semitransparent portion or the opaque portion, the vibration apparatus **500** can be configured to be semitransparent or opaque and can be disposed at the semitransparent portion or the opaque portion of the vehicle glass window **230**. For example, the vibration apparatus **500** can be a transparent sound generating module.

[0256] Referring to FIGS. **1** to **6** in conjunction with FIG. **10**, the vibration apparatus **500** can be disposed at one surface (or an indoor surface) of the vehicle glass window **230** exposed at inside or an indoor space IS of the vehicular apparatus or the vehicle **10**. For example, a sound apparatus **30** for vehicles can include at least one or more vibration apparatus **500** disposed at the vehicle glass window **230** or a plurality of the vibration apparatus **500** disposed at the vehicle glass window **230**. As an embodiment of the present disclosure, the vibration apparatus **500** can be disposed at at least one or more of the front glass window and the side glass window and can be further disposed at at least one or more of the rear glass window and the roof glass window.

[0257] The at least one or more of the thirteenth and fourteenth sound generating modules **31**N and **31**O can include the transparent or semitransparent vibration apparatus **500**. For example, when the at least one or more of the thirteenth and fourteenth sound generating modules **31**N and **31**O includes the vibration apparatus **500**, the at least one or more of the thirteenth and fourteenth sound generating modules **31**N and **31**O can vibrate the vehicle glass window **230** to output a sound based on a vibration of the vehicle glass window **230**. For example, the at least one or more of the thirteenth and fourteenth sound generating modules **31**N and **31**O can directly vibrate the vehicle glass window **230** to output a sound based on a vibration of the vehicle glass window **230**.

[0258] According to an embodiment of the present disclosure, the vehicle glass window **230** can include the front glass window, the side glass window **230**B, and the rear glass window **230**C. According to an embodiment of the present disclosure, the vehicle glass window **230** can further include the roof glass window. For example, when the vehicular apparatus or the vehicle **10** includes the roof glass window, a portion of a region of the roof frame and the roof interior material **130**C can be replaced with the roof glass window. For example, when the vehicular apparatus or the vehicle **10** includes the roof glass window, the third sound generating module **31**C can be configured to vibrate a periphery portion of the roof interior material **130**C surrounding the roof glass window.

[0259] Referring to FIGS. **11** to **13**, the thirteenth sound generating module **31**N according to an embodiment of the present disclosure can be disposed at the side glass window **230**B and can be configured to output a sound by vibrating itself (or self-vibration thereof), or can be configured to indirectly or directly vibrate the side glass window **230**B to output a sound based on a vibration of the side glass window **230**B. For example, the thirteenth sound generating module **31**N can directly vibrate the side glass window **230**B to output a sound based on the vibration of the side glass window **230**B.

[0260] According to an embodiment of the present disclosure, the side glass window **230**B can include a first side glass window (or a right front window) **230**B**2**, a second side glass window (or a left rear window) **230**B**3**, a third side glass window (or a right rear window) **230**B**4**, and a fourth side glass window (or a left front window).

[0261] According to an embodiment of the present disclosure, the thirteenth sound generating module **31**N can be disposed at at least one or more of the first to third side glass windows **230**B2 to **230**B**4**. For example, at least one or more of the first to third side glass windows **230**B**2** to **230**B**4** can include at least one or more thirteenth sound generating module **31**N.

[0262] For example, the thirteenth sound generating module **31**N can be disposed at at least one or more of the first to third side glass windows **230**B**2** to **230**B**4** and can output a sound by vibrating itself (or self-vibration thereof), or can vibrate a corresponding side glass windows **230**B2 to **230**B4 to output a sound. For example, the thirteenth sound generating module **31**N can directly vibrate the side glass windows **230**B**2** to **230**B**4** to output the sound. For example, the thirteenth sound generating module **31**N can be configured to output the sound of 150 Hz to 20 kHz. For example, the thirteenth sound generating module **31**N disposed at at least one or more of the first to third side glass windows **230**B**2** to **230**B**4** can have a same sound output characteristic or different sound output characteristics. For example, the thirteenth sound generating module **31**N disposed at at least one or more of the first to third side glass windows **230**B**2** to **230**B**4** can be configured to output the sound of 150 Hz to 20 kHz. For example, the thirteenth sound generating module **31**N can be a side window speaker.

[0263] Referring to FIG. **11**, the fourteenth sound generating module **31**O according to an embodiment of the present disclosure can be disposed at the rear glass window **230**C and can output a sound by vibrating itself (or self-vibration thereof), or can vibrate the rear glass window **230**C to output a sound based on a vibration of the rear glass window **230**C. For example, the fourteenth sound generating module **31**O can directly vibrate the rear glass window **230**C to output the sound based on the vibration of the rear glass window **230**C.

[0264] According to an embodiment of the present disclosure, the rear glass window **230**C can include a first region corresponding to a rear portion of the first rear seat BS**1**, a second region corresponding to a rear portion of the second rear seat BS**2**, and a third region corresponding to a rear portion of the third rear seat BS**3**. According to an embodiment of the present disclosure, the fourteenth sound generating module **31**O can be disposed at at least one or more of the first and second regions of the rear glass window **230**C. For example, the fourteenth sound generating module **31**O can be disposed at at least one or more of the first to third regions of the rear glass window **230**C. For example, the fourteenth sound generating module **31**O can be disposed at each of the first and second regions of the rear glass window **230**C, or can be disposed at each of the first to third regions of the rear glass window **230**C. For example, the fourteenth sound generating module **31**O can be disposed at at least one or more of the first and second regions of the rear glass window **230**C, or can be disposed at at least one or more of the first to third regions of the rear glass window **230**C. For example, the fourteenth sound generating module **31**O can be configured to output a sound of about 150 Hz to about 20 kHz. For example, the fourteenth sound generating module **31**O disposed at at least one or more of the first to third regions of the rear glass window **230**C can have a same sound output characteristic or different sound output characteristics. For example, the fourteenth sound generating module **31**O disposed at each of the first to third regions of the rear glass window **230**C can have a same sound output characteristic or different sound output characteristics. For example, the fourteenth sound generating module **31**O disposed at at least one or more of the first and second regions of the rear glass window **230**C can be configured to output a sound of about 150 Hz to about 20 kHz, or can be one or more of a woofer, a mid-woofer, and a sub-woofer. For example, the fourteenth sound generating module **31**O can be a rear window speaker.

[0265] Referring to FIGS. **10** and **11**, the vehicular apparatus or the vehicle **10** according to an embodiment of the present disclosure can further include a woofer speaker WS which is disposed at at least one or more of a dashboard **130**A, a door frame, and a rear package interior material **130**J.

[0266] The woofer speaker WS according to an embodiment of the present disclosure can include at least one or more of a woofer, a mid-woofer, and a sub-woofer. For example, the woofer speaker WS can be a speaker which outputs a sound of about 60 Hz to about 150 Hz. Therefore, the woofer speaker WS can output a sound of about 60 Hz to about 150 Hz, and thus, can enhance a low-pitched sound band characteristic of a sound which is output to an indoor space.

[0267] According to an embodiment of the present disclosure, the woofer speaker WS can be disposed at at least one or more of first and second regions of the dashboard **130**A. According to an embodiment of the present disclosure, the woofer speaker WS can be disposed at each of first to fourth door frames of the door frame and can be exposed at a lower region of each of the first to fourth door interior materials **130**D1 to **130**D4 of the door interior material **130**D. For example, the woofer speaker WS can be disposed at at least one or more of the first to fourth door frames of the door frame and can be exposed at the lower regions of at least one or more of the first to fourth door interior materials **130**D**1** to **130**D**4** of the door interior material **130**D. According to an embodiment of the present disclosure, the woofer speaker WS can be disposed at at least one or more of the first and second regions of the rear package interior material **130**J. For example, the fourth sound generating module **31**D disposed at the lower region of at least one or more of the first to fourth door interior materials **130**D**1** to **130**D**4** can be replaced by the woofer speaker WS.

[0268] The vehicular apparatus according to an embodiment of the present disclosure can further include a garnish member and one or more sound generation apparatus **30** arranged in the vehicle interior material **130**. For example, one or more sound generation apparatus **30** can be placed on the garnish member and the vehicle interior material **130** to output sound. For example, at least one of the garnish member and the vehicle interior material **130** can output sound according to the vibration of one or more sound generation apparatus (or vibration apparatus).

[0269] The garnish member can be configured to cover a portion of the door interior material **130**D exposed at the indoor space, but embodiments of the present disclosure are not limited thereto. For example, the garnish member can be configured to cover a portion of one or more of the dashboard **130**A, the pillar interior material **130**B, and the roof interior material **130**C, which are exposed at the indoor space.

[0270] The garnish member according to an embodiment of the present disclosure can include a metal material or a nonmetal material (or a composite nonmetal material) having a material characteristic suitable for generating a sound based on a vibration. For example, a metal material of the garnish member can include any one or more materials of stainless steel, aluminum (Al), an Al alloy, a magnesium (Mg), a Mg alloy, and a magnesium-lithium (Mg—Li) alloy, but embodiments of the present disclosure are not limited thereto. The nonmetal material (or the composite nonmetal material) of the garnish member can include one or more of wood, plastic, glass, cloth, fiber, rubber, paper, carbon, and leather, but embodiments of the present disclosure are not limited thereto. For example, the garnish member can include a metal material having a material characteristic suitable for generating a sound of a high-pitched sound band, but embodiments of the present disclosure are not limited thereto. For example, the high-pitched sound band can have a frequency of 1 kHz or more or 3 kHz or more, but embodiments of the present disclosure are not limited thereto.

[0271] As noted above in connection with FIGS. **10** to **13**, the vibration apparatus **500** may be incorporated into a body of the vehicular apparatus. The plurality of vehicle interior materials can also be coupled to the body of the vehicular apparatus. In some embodiments, the vehicle interior materials are additional materials removably attached or coupled to the body of the vehicular apparatus. In other embodiments, the vehicle interior materials are integrated into the body, thereby forming part of the body.

[0272] In some cases, the vibration apparatus is coupled to either the body or the vehicle interior materials of the vehicular apparatus. Accordingly, in certain cases, the vibration apparatus, under normal operation, is not separated or detached from the body. However, it may be separated in instances where the vehicular apparatus needs repair.

[0273] The body of the vehicular apparatus includes a motor mounted to the body. The motor may include a combustion engine, an electric motor, or a hybrid system combining an internal combustion engine (usually fueled by gasoline or diesel) with an electric motor, or the like. Accordingly, in some embodiments, the vehicular apparatus may include not only conventional fuel vehicles but also electric vehicles or other vehicles that run on clean energy.

[0274] One or more sound generating apparatus **30** can be placed between the garnish member and the vehicle interior material **130**. For example, the one or more sound generation apparatus **30** can be garnish speakers, etc., but the embodiments of the present disclosure are not limited thereto.

[0275] One or more sound generation apparatus **30** according to the embodiments of the present disclosure can include the vibration apparatus **500** described with reference to FIGS. **1** to **7**B. The one or more sound generation apparatus **30** can be placed on the main interior material or the vehicle interior material **130**, and the garnish member, and can be connected or coupled to the garnish member.

[0276] One or more sound generation apparatus **30** according to the embodiments of the present disclosure can be configured to indirectly or directly vibrate the garnish member to output sound into the interior space of the transportation device. For example, the one or more sound generation apparatus **30** can be configured to output high-pitched sound, but the embodiments of the present disclosure are not limited thereto.

[0277] The vehicular apparatus or the vehicle **10** according to an embodiment of the present disclosure can output a sound to the indoor space through at least one or more of the first sound generating apparatus **30-1** disposed at the vehicle interior material **130**, the second sound generating apparatus **30-2** disposed at the vehicle interior material **130** exposed at the indoor space, the third sound generating apparatus **30-3** disposed at the vehicle glass window **230**, and the fourth sound generating apparatus disposed at the garnish member, and thus, can output the sound by a vehicle interior material **130** as a vibration plate or a sound vibration plate, thereby outputting a multichannel surround stereo sound.

[0278] A vibration apparatus and vehicular apparatus comprising same according to example embodiments of the present disclosure are described below.

[0279] A vibration apparatus according to one or more embodiments of the present disclosure can comprises a cover member having first and second areas, a vibration part in each of the first and second areas, and a support member between the cover member and the vibration part.

[0280] According to one or more embodiments of the present disclosure, the vibration part can include a first vibration part in the first area, and a plurality of second vibration parts in the second area.

[0281] According to one or more embodiments of the present disclosure, the first area can be smaller than the second area in size.

[0282] According to one or more embodiments of the present disclosure, each of the plurality of second vibration parts can be smaller than the first vibration part in size.

[0283] According to one or more embodiments of the present disclosure, each of the first vibration part and the plurality of second vibration parts can include a vibration layer including a piezoelectric material, a first electrode layer on a first surface of the vibration layer, and a second electrode layer on a second surface different from the first surface of the vibration layer. The vibration layers of the first vibration part and the second vibration part can have different sizes.

[0284] According to one or more embodiments of the present disclosure, the vibration layer of the second vibration part can have a size smaller than that of the vibration layer of the first vibration part.

[0285] According to one or more embodiments of the present disclosure, the first electrode layer of the first vibration part and the first electrode layer of the second vibration part can be electrically connected to each other.

[0286] According to one or more embodiments of the present disclosure, the support member can include a first support member between the cover member and the first vibration part, and a second support member between the cover member and the second vibration part. The first support member and the second support member can include the same material.

[0287] According to one or more embodiments of the present disclosure, the first support member and the second support member can have different thicknesses.

[0288] According to one or more embodiments of the present disclosure, the vibration apparatus can further comprise a flexible part configured between the first vibration part and the second vibration part and between each of the plurality of second vibration parts. The flexible part can be on a rear surface of the first support member.

[0289] According to one or more embodiments of the present disclosure, the cover member can further include a third area parallel to the second area, and the first to third areas can have the same length.

[0290] According to one or more embodiments of the present disclosure, the vibration part can further include a third vibration part in the third area.

[0291] According to one or more embodiments of the present disclosure, each of the first to third vibration parts can include a vibration layer including a piezoelectric material, a first electrode layer on a first surface of the vibration layer, and a second electrode layer on a second surface different from the first surface of the vibration layer. The vibration layers of the first vibration part and the third vibration part can have the same size, and the vibration layers of the first vibration part and the second vibration part can have different sizes.

[0292] According to one or more embodiments of the present disclosure, the vibration layer of the first vibration part can be larger than the vibration layer of the second vibration part in size.

[0293] According to one or more embodiments of the present disclosure, the first electrode layer of the first vibration part can be electrically connected to the first electrode layer of the second vibration part, and the first electrode layer of the second vibration part and the first electrode layer of the third vibration part can be electrically connected to each other.

[0294] According to one or more embodiments of the present disclosure, the support member can include a first support member between the cover member and the first vibration part, a second support member between the cover member and the plurality of second vibration parts, and a third support member between the cover member and the third vibration part. The first to third support members can include the same material.

[0295] According to one or more embodiments of the present disclosure, the first support member and the second support member can have different thicknesses, and the first support member and the third support member can have the same thickness.

[0296] According to one or more embodiments of the present disclosure, the vibration apparatus can further comprise a flexible part between the first vibration part and the second vibration part and between the second vibration part and the third vibration part. The flexible part can be disposed on a rear surface of the first support member and a rear surface of the third support member.

[0297] According to one or more embodiments of the present disclosure, the support member can include a first support member between the cover member and the first vibration part, a second support member between the cover member and the plurality of second vibration parts, and a third support member between the cover member and the third vibration part. The first and second support members can include different materials, and the first and third support members can include the same material.

[0298] According to one or more embodiments of the present disclosure, the first to third support members can have the same thickness.

[0299] According to one or more embodiments of the present disclosure, each of the first to third support members can include a support plate, and a protrusion vertically protruding from facing sides of the support plate.

[0300] According to one or more embodiments of the present disclosure, the support plates of each of the first support member and the second support member can have different thicknesses, and the support plates of the first support member and the third support member can have the same thickness.

[0301] According to one or more embodiments of the present disclosure, the protrusions of the first support member and the second support member can have different thicknesses, and the protrusions of the first support member and the third support member can have the same thickness.

[0302] According to one or more embodiments of the present disclosure, a thickness of the support plate of the first support member can be smaller than a thickness of the support plate of the second support member, and a thickness of the protrusion of the first support member can be greater than a thickness of the protrusion of the second support member.

[0303] According to one or more embodiments of the present disclosure, vibration apparatus can further comprise a flexible part between the first vibration part and the second vibration part adjacent to each other and between the second vibration part and the third vibration part. The flexible part can be disposed on a rear surface of the support plate in each of the first support member and the third support member.

[0304] According to one or more embodiments of the present disclosure, the support member can include at least any one of aluminum Al, copper Cu, and plastic.

[0305] According to one or more embodiments of the present disclosure, the thickness of the support member can be 0.5 to 2.0 times the thickness of the vibration layer.

[0306] According to one or more embodiments of the present disclosure, the cover member can include a first cover member on a first surface of the vibration part, and a second cover member on a second surface different from the first surface of the vibration part. The support member can be between the vibration part and the first cover member.

[0307] According to one or more embodiments of the present disclosure, the vibration apparatus can further comprise a signal cable connected to the vibration part. The signal cable can include a first signal line between the vibration part and the first cover member and a second signal line between the vibration part and the second cover member.

[0308] A vehicular apparatus according to one or more embodiments of the present disclosure can comprises an interior material exposed to an indoor space, and at least one sound generating apparatus configured to output sound to the indoor space. The at least one sound generating apparatus can include a vibration apparatus comprising a cover member having first and second areas, a vibration part in each of the first and second areas, and a support member between the cover member and the vibration part.

[0309] According to one or more embodiments of the present disclosure, the interior material can include one or more among metal, wood, rubber, plastic, carbon, glass, fiber, cloth, paper, mirror, and leather.

[0310] According to one or more embodiments of the present disclosure, the interior material can include at least one of a dashboard, a pillar interior material, a roof interior material, a door interior material, a seat interior material, a handle interior material, a floor interior material, a rear package interior material, an overhead console, a rear view mirror, a glove box, a garnish member, and a sun visor, and the at least one sound generating apparatus can generate sound by vibrating at least one of the dashboard, the pillar interior material, the roof interior material, the door interior material, the seat interior material, the handle interior material, the floor interior material, the rear package interior material, the overhead console, the rear view mirror, the glove box, the garnish member, and the sun visor.

[0311] A vibration apparatus according to one or more embodiments of the present disclosure can be applied to or included in a vibration apparatus disposed in an apparatus. The apparatus according to one or more example embodiments of the present disclosure can be applied to or included in mobile apparatuses, video phones, smart watches, watch phones, wearable apparatuses, foldable apparatuses, rollable apparatuses, bendable apparatuses, flexible apparatuses, curved apparatuses, sliding apparatuses, variable apparatuses, electronic organizers, electronic books, portable multimedia players (PMPs), personal digital assistants (PDAs), MP3 players, mobile medical devices, desktop personal computers (PCs), laptop PCs, netbook computers, workstations, navigation apparatuses, automotive navigation apparatuses, automotive display apparatuses, automotive apparatuses, theatre apparatuses, theatre display apparatuses, TVs, wall paper display apparatuses, signage apparatuses, game machines, notebook computers, monitors, cameras, camcorders, and home appliances, or the like. In addition, the vibration apparatus according to some example embodiments of the present disclosure can be applied to or included in organic light-emitting lighting apparatuses or inorganic light-emitting lighting apparatuses. When the vibration apparatus according to one or more example embodiments of the present disclosure is applied to or included in lighting apparatuses, the vibration apparatus can act as a lighting device and a speaker.

[0312] In addition, when the vibration apparatus according to some example embodiments of the present disclosure is applied to or included in a mobile device, or the like, the vibration apparatus can be one or more of a speaker, a receiver, and a haptic device, but embodiments of the present disclosure are not limited thereto.

[0313] The various embodiments described above can be combined to provide further embodiments. All of the U.S. patents, U.S. patent application publications, U.S. patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

[0314] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

1. A vibration apparatus comprising:

a cover member having first and second areas;

a vibration part in each of the first and second areas; and

a support member between the cover member and the vibration part.

2. The vibration apparatus according to claim **1**,

wherein the vibration part includes:

a first vibration part in the first area; and

a plurality of second vibration parts in the second area.

3. The vibration apparatus according to claim **1**, wherein the first area is smaller than the second area in size.

4. The vibration apparatus according to claim **2**, wherein each of the plurality of second vibration parts is smaller than the first vibration part in size.

5. The vibration apparatus according to claim **2**,

wherein each of the first vibration part and the plurality of second vibration parts includes:

a vibration layer including a piezoelectric material, the vibration layer having a first surface and a second surface different from the first surface;

a first electrode layer on the first surface of the vibration layer; and

a second electrode layer on the second surface of the vibration layer,

wherein the vibration layers of the first vibration part and the second vibration part have different sizes.

6. The vibration apparatus according to claim **5**, wherein the vibration layer of the second vibration part has a size smaller than that of the vibration layer of the first vibration part.

7. The vibration apparatus according to claim **5**, wherein the first electrode layer of the first vibration part and the first electrode layer of the second vibration part are electrically connected to each other.

8. The vibration apparatus according to claim **2**,

wherein the support member includes:

a first support member between the cover member and the first vibration part, the first support member having a first surface; and

a second support member between the cover member and the second vibration part,

wherein the first support member and the second support member include a same material.

9. The vibration apparatus according to claim **8**, wherein the first support member and the second support member have different thicknesses.

10. The vibration apparatus according to claim **8**, further comprising:

a flexible part configured between the first vibration part and the second vibration portion and between each of the plurality of second vibration parts,

wherein the flexible part is on the first surface of the first support member.

11. The vibration apparatus according to claim **2**,

wherein the cover member further includes a third area parallel to the second area, and

wherein the first to third areas have a same length from a plan view.

12. The vibration apparatus according to claim **11**, wherein the vibration part further includes a third vibration part in the third area.

13. The vibration apparatus according to claim **12**,

wherein each of the first to third vibration parts includes:

a vibration layer including a piezoelectric material;

a first electrode layer on a first surface of the vibration layer; and

a second electrode layer on a second surface different from the first surface of the vibration layer, wherein the vibration layers of the first vibration part and the third vibration part have a same size, and

the vibration layers of the first vibration part and the second vibration part have different sizes.

14. The vibration apparatus according to claim **13**, wherein the vibration layer of the first vibration part is larger than the vibration layer of the second vibration part in size.

15. The vibration apparatus according to claim **13**,

wherein the first electrode layer of the first vibration part is electrically connected to the first electrode layer of the second vibration part, and

wherein the first electrode layer of the second vibration part and the first electrode layer of the third vibration part are electrically connected to each other.

16. The vibration apparatus according to claim **12**,

wherein the support member includes:

a first support member between the cover member and the first vibration part, the first support member having a first surface;

a second support member between the cover member and the plurality of second vibration parts; and

a third support member between the cover member and the third vibration part, the third support member having a second surface,

wherein the first to third support members include a same material.

17. The vibration apparatus according to claim **16**,

wherein the first support member and the second support member have different thicknesses, and

wherein the first support member and the third support member have the same thickness.

18. The vibration apparatus according to claim **12**,

wherein the support member includes:

a first support member between the cover member and the first vibration part;

a second support member between the cover member and the plurality of second vibration parts; and

a third support member between the cover member and the third vibration part,

wherein the first and second support members include different materials,

wherein the first and third support members include a same material.

19. A vehicular apparatus comprising:

a body having a motor mounted therein;

a plurality of vehicle interior materials coupled to the body; and

a vibration apparatus coupled to at least one vehicle interior material of the plurality of vehicle interior materials, the vibration apparatus including:

a cover member having first and second areas;

a vibration part in each of the first and second areas, the vibration part including:

a first vibration part in the first area; and

a plurality of second vibration parts in the second area; and

a support member between the cover member and the vibration part.

wherein each of the plurality of second vibration parts is smaller than the first vibration part in size.

20. The vehicular apparatus according to claim **19**,

wherein each of the first vibration part and the plurality of second vibration parts includes:

a vibration layer including a piezoelectric material, the vibration layer having a first surface and a second surface different from the first surface;

a first electrode layer on the first surface of the vibration layer; and

a second electrode layer on the second surface of the vibration layer,

wherein the vibration layers of the first vibration part and the second vibration part have different sizes.

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